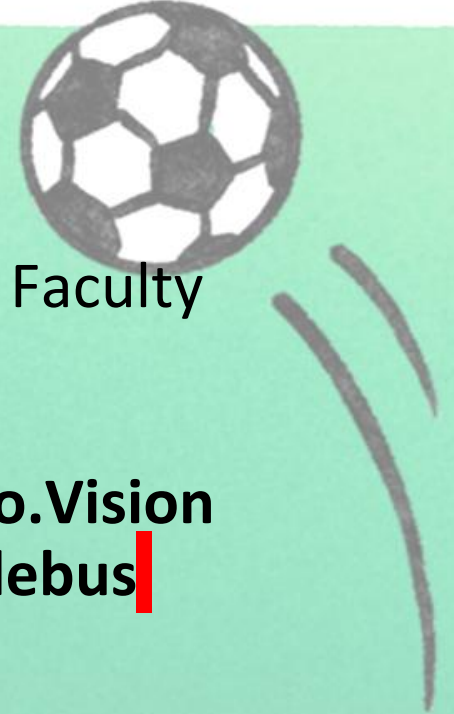




WELCOME EVERYONE TO WEC 2026!

Agenda for today

- 9:00 – General introduction by **Piotr Wójcik**
- 9:10 – Warm welcome by **Gabriela Grotkowska**, Dean of our Faculty
- 9:20 – Description of the contest rules by **Jacek Lewkowicz**
- 9:30 – Guest Lecture by **Piotr Żmijewski** from **Legia LAB**
- 9:45 – Insight lecture by **Radosław Marcinkowski** from **ReSpo.Vision**
- 10:00 – Competition strategy recommendation by **Marcin Chlebus**
- 10:10 – Description of the contest problem and presentation of the data by **Piotr Wójcik**
- 10:30 – The 24-hour **competition begins!**
- 10:30 – 11:30 – Time for potential **confidential** questions sent via email to wec@wne.uw.edu.pl





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WARSAW *econometric* **CHALLENGE**

R·V ReSpo·Vision

GENERAL INTRODUCTION

by Piotr Wójcik

Organizing Committee



- **Marcin Chlebus, PhD**
- Department of Data Science



- **Rafał Woźniak, PhD**
- Department of Statistics and Econometrics



- **Jacek Lewkowicz, prof. UW**
- Department of Political Economy



- **Piotr Wójcik, prof. UW**
- Department of Data Science



ReSpo.Vision & Legia LAB representatives



- **Radosław Marcinkowski**
- Sales Director & Key Account Manager



- **dr hab. Piotr Żmijewski**
- Head of Legia LAB



Administrative and IT support



Krystyna
Paczoska



Monika
Oponowicz



Igor
Nowicki



Ireneusz
Kulesz



Dominika
Huczek

Participants

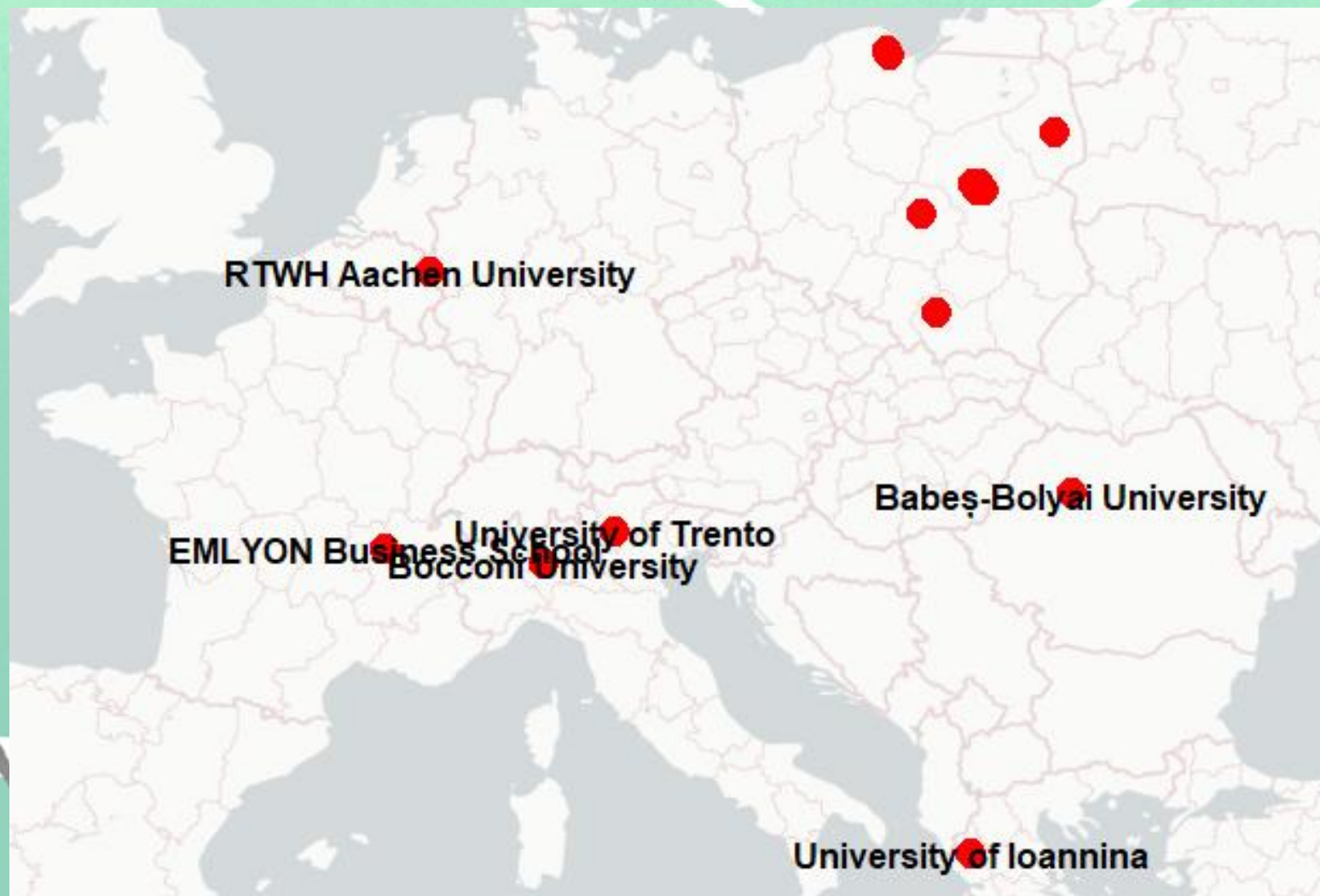
- 1st edition (2021)
 - local at the Faculty of Economic Sciences
 - 5 teams, 16 participants
- 2nd edition (2022) - co-organized by LOT Polish Airlines
 - students from 6 Polish universities
 - 28 teams, 93 participants
- 3rd edition (2023) - co-organized by the High Tech Foundation
 - students from 23 countries representing 21 European universities
 - 46 teams, 134 participants
- 4th edition (2024)
 - students from 19 countries representing 21 European and 1 US university
 - 36 teams, 107 participants
- 5th edition (2025) - co-organized by Dataplace.ai
 - students from 24 countries representing 19 European universities
 - 45 teams, 116 participants

6th edition (2026) co-organized by ReSpo.Vision

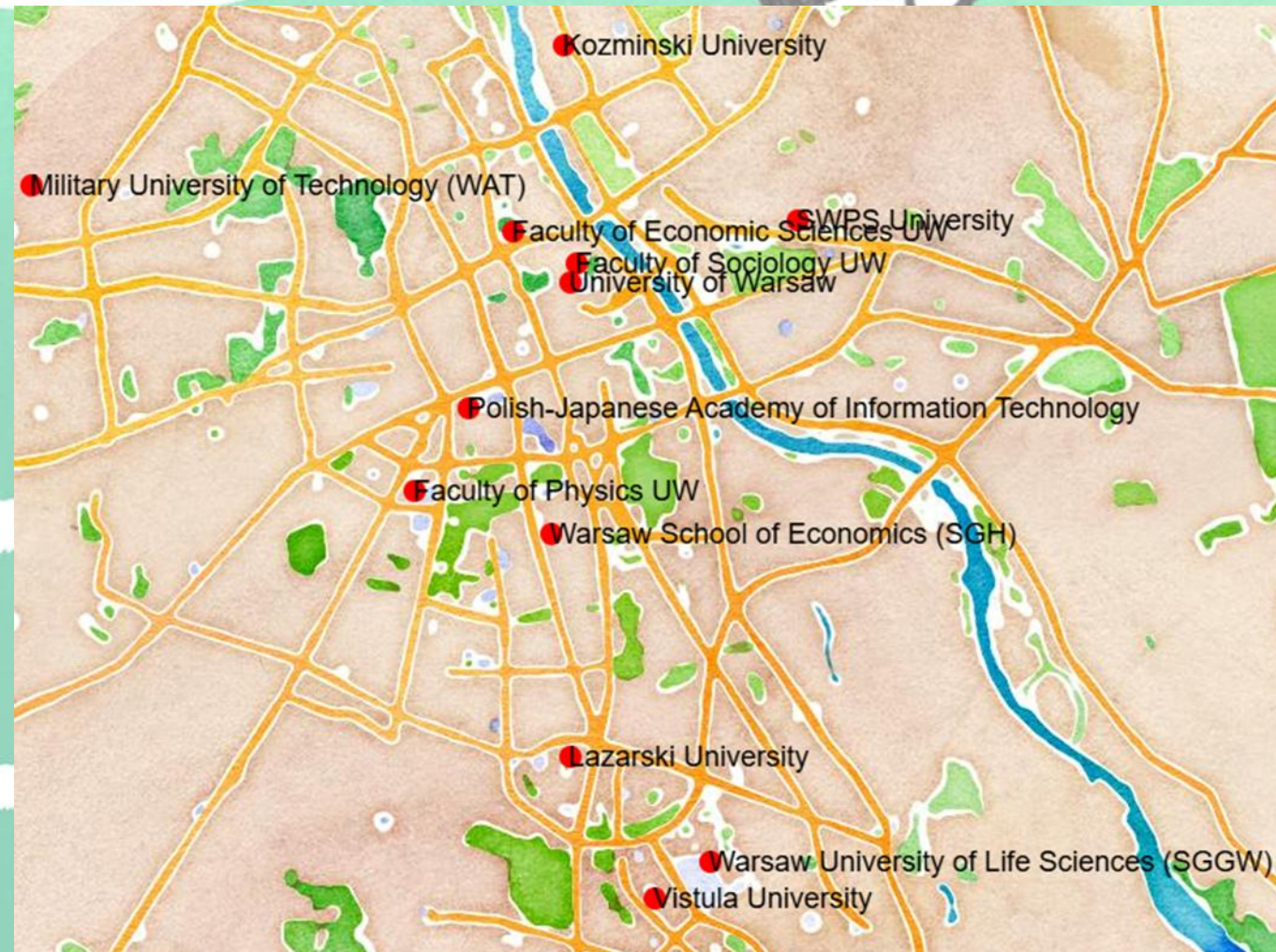
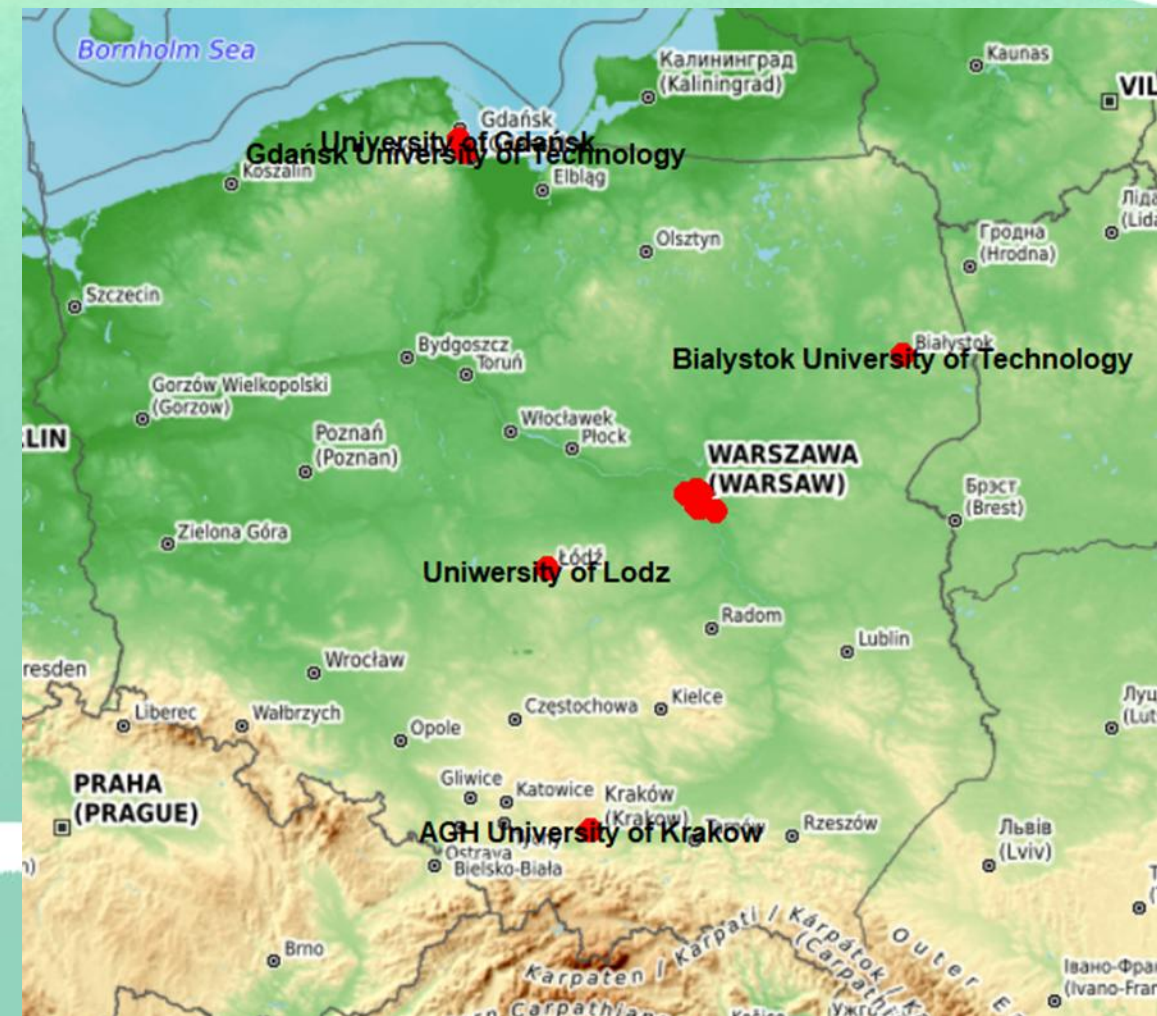
- **students from 20 countries representing 21 European universities**
- **118 participants**
- **46 teams**



Participants 2026 – all universities



Participants 2026 – Polish universities





WARM WELCOME
by Gabriela Grotkowska,
Dean of the Faculty of Economic Sciences,
University of Warsaw



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WARSAW *econometric* **CHALLENGE**

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**DESCRIPTION
OF THE CONTEST RULES**
by Jacek Lewkowicz

What we expect from you



Your **core goal** is to prepare a **research paper** in which you describe your solution to the task

- Present some background and brief literature review
- Define research questions/hypotheses
- Describe the data
- Comment on the methodology
- Present your results
- Draw conclusions and provide discussion

Criteria for evaluation

adequacy

comprehensivity

interpretability

predictive power



How to submit your works?

- Use a **form** available on the website <https://wec.wne.uw.edu.pl/>
- Do it before **16:00** CET on Sunday, April 26th 2026
- Submit the **article**, full **codes** and the **final dataset** used, so that we can **reproduce** your analyses
- Consider sharing with us also some **additional deliverables** (details in a moment)



Prizes

- Satisfaction and glory – priceless 😊
- Prizes (for each team member)
 - 1st place: PLN 2000 (ca. EUR 470)
 - 2nd place: PLN 1000 (ca. EUR 235)
 - 3rd place: PLN 500 (ca. EUR 118)





Determinants of running performance in football

by Piotr Żmijewski

Among the 55 UEFA leagues, 7 of the highest-ranked leagues feature in the global top 10 for high-intensity running performance

Ostatnia aktualizacja: 21.05.2025 22:58:30

RANKING KRAJOWY UEFA 2025

Poz.	Kraj	2020/2021	2021/2022	2022/2023	2023/2024	2024/2025	Razem	Kluby
1	Anglia	24,357	21,000	23,000	17,375	29,178	114,910	1/7
2	Włochy	16,285	15,714	22,357	21,000	21,875	97,231	1/8
3	Hiszpania	19,500	18,428	16,571	16,062	23,892	94,453	1/7
4	Niemcy	15,214	16,214	17,125	19,357	18,421	86,331	0/8
5	Francja	7,916	18,416	12,583	16,250	17,642	72,807	1/7
6	Holandia	9,200	19,200	13,500	10,000	15,250	67,150	0/6
7	Portugalia	9,600	12,916	12,500	11,000	16,250	62,266	0/5
8	Belgia	6,000	6,600	14,200	14,400	15,650	56,850	0/5
9	Czechy	6,600	6,700	6,750	13,500	10,550	44,100	0/5
10	Turcja	3,100	6,700	11,800	12,000	10,300	43,900	0/5
11	Norwegia	6,500	7,625	5,750	8,000	11,812	39,687	0/4
12	Grecja	5,100	8,000	2,125	11,400	12,687	39,312	0/4
13	Austria	6,700	10,400	4,900	4,800	9,650	36,450	0/5
14	Szkocja	8,500	7,900	3,500	6,400	9,250	35,550	0/5
15	Polska	4,000	4,625	7,750	6,875	11,750	35,000	0/4
16	Dania	4,125	7,800	5,900	8,500	7,656	33,981	0/4
17	Szwajcaria	5,125	7,750	8,500	5,200	7,050	33,625	0/5
18	Izrael	7,000	6,750	6,250	8,750	2,875	31,625	0/4
19	Cypr	4,000	4,125	5,100	3,750	10,562	27,537	0/4
20	Szwecja	2,500	5,125	6,250	1,875	11,375	27,125	0/4

Fast runs index, 30 leagues worldwide

Domestic league matches, current or last completed season until 08/04/2024. Data: SkillCorner.

Index		Sprints	Speed	Accelerations
100.0	Premier League (ENG)	100.0	100.0	99.9
85.5	Championship (ENG/2)	83.8	78.9	100.0
75.1	Serie A (ITA)	81.1	79.5	73.2
72.3	Eredivisie (NED)	77.4	69.3	80.7
70.4	Super League (SUI)	81.8	78.8	60.1
69.9	Ligue 2 (FRA/2)	84.5	57.1	81.2
68.2	Primeira Liga (POR)	83.8	60.7	72.9
67.7	Ligue 1 (FRA)	89.3	72.9	52.1
67.0	Brasileirão (BRA)	47.9	87.3	74.2
65.5	La Liga (ESP)	79.5	66.2	63.2
62.3	Pro League (BEL)	84.4	61.5	55.1
62.0	Super League 1 (GRE)	64.3	58.2	77.8
60.2	Süper Lig (TUR)	68.4	65.1	60.8
58.5	Superliga (DEN)	77.6	59.6	53.5
56.0	Bundesliga (GER)	73.5	64.0	45.5
54.7	Bundesliga (AUT)	65.2	62.1	52.3
54.3	La Liga 2 (ESP/2)	65.9	54.2	59.5
54.2	Premiership (SCO)	50.5	66.0	60.7
51.0	Czech Liga (CZE)	66.9	67.9	33.3
49.4	A-League Men (AUS)	75.3	59.3	30.9

Establishing reference values for the match running performances of thirteen specific positional roles at UEFA Euro 2024

AUTHORS: Shuyao Chen¹, Piotr Zmijewski^{2,3}, Paul S Bradley⁴

🎯 1. Research Objective

- Establishing normative reference values for physical performance metrics based on UEFA Euro 2024 match data.
- Quantifying locomotive demands across distinct game phases and temporal intervals (e.g., peak 1-minute intensity periods).
- Integrating tactical, technical, and individual performance context specific to each playing position.



Establishing reference values for the match running performances of thirteen specific positional roles at UEFA Euro 2024

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AUTHORS: Shuyao Chen¹, Piotr Zmijewski^{2,3}, Paul S Bradley⁴

† Corresponding author:

2. Data and Methods

✦ **Tournament:** UEFA Euro 2024 (51 matches, 24 national teams).

👤 **sample size:** 544 Full-match appearances (90+ min).

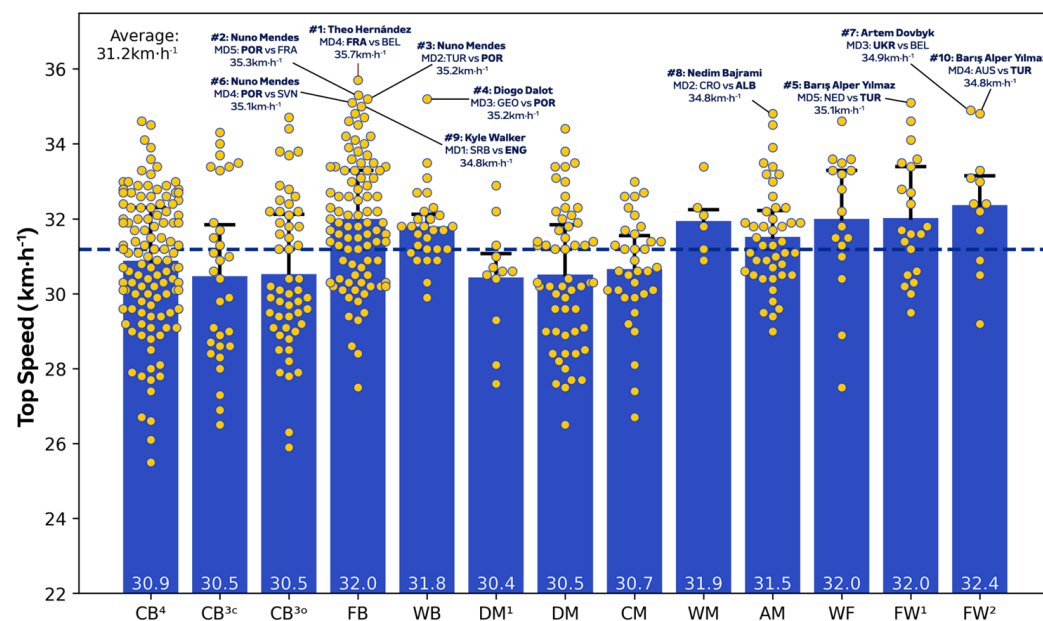
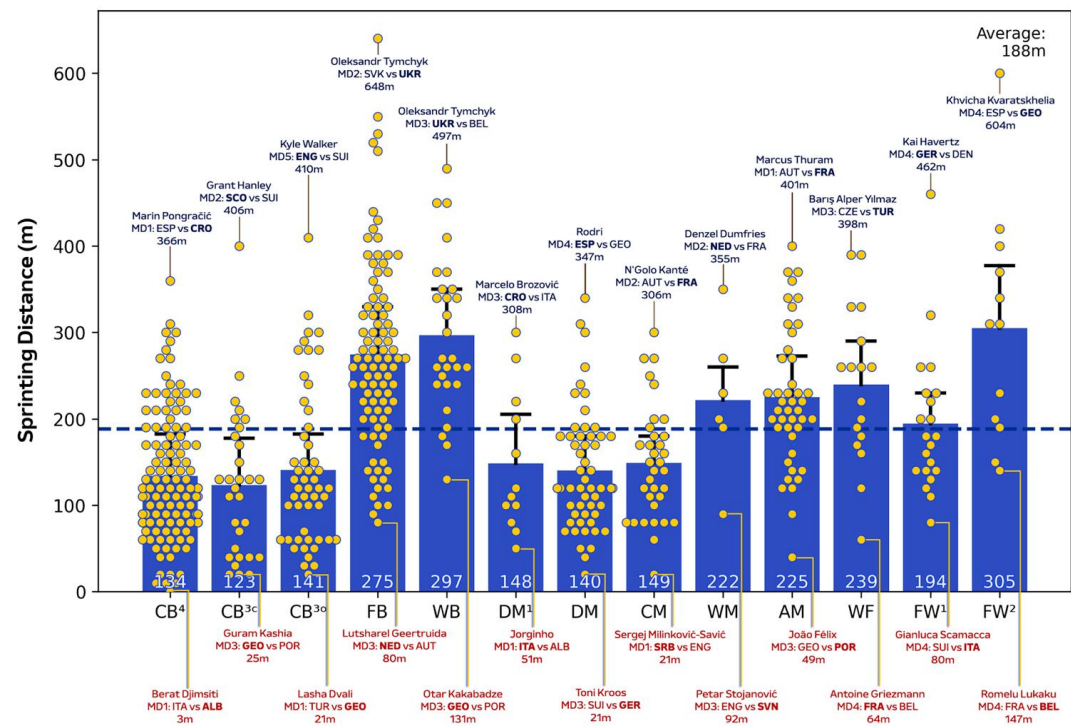
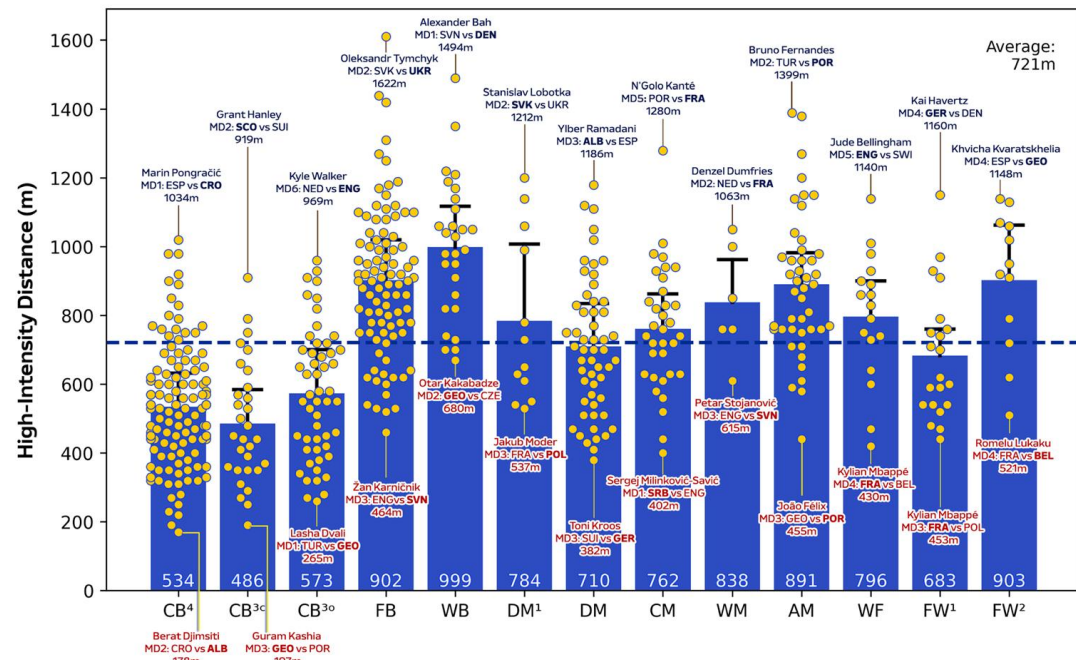
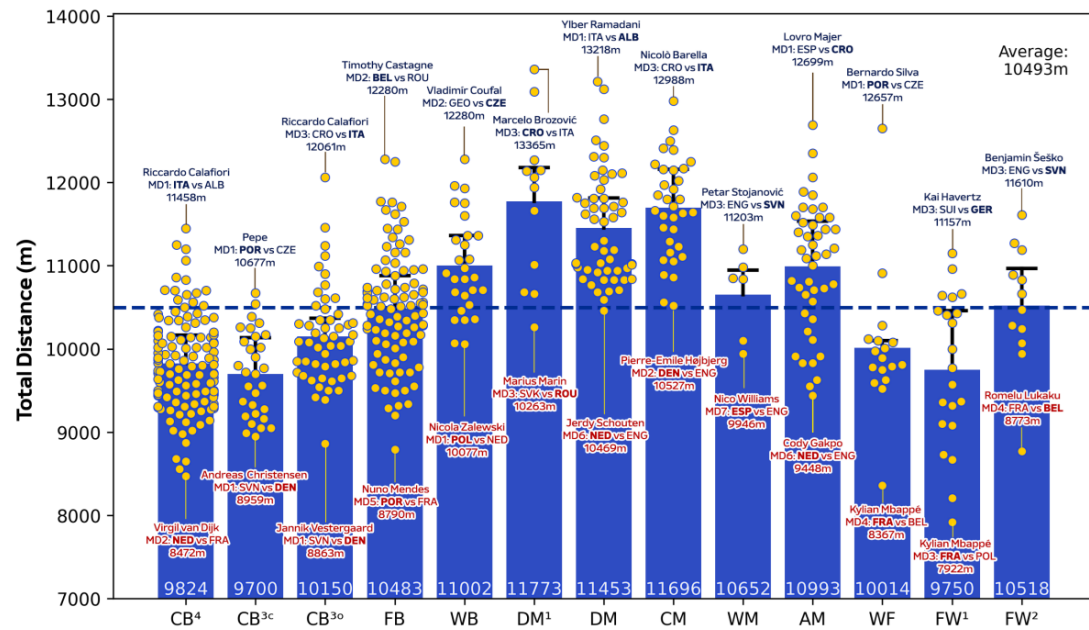
❓ **Tracking system:** optic tracking *Hawk-Eye Innovations Ltd.*

📊 **Locomotion Speed:**

- Standing: 0–0.9 km/h
- Walking: 1–6.9 km/h
- Jogging: 7–14.9 km/h
- Running: 15–19.9 km/h
- High-Speed Running (HSR): 20–24.9 km/h
- Sprinting: ≥ 25 km/h

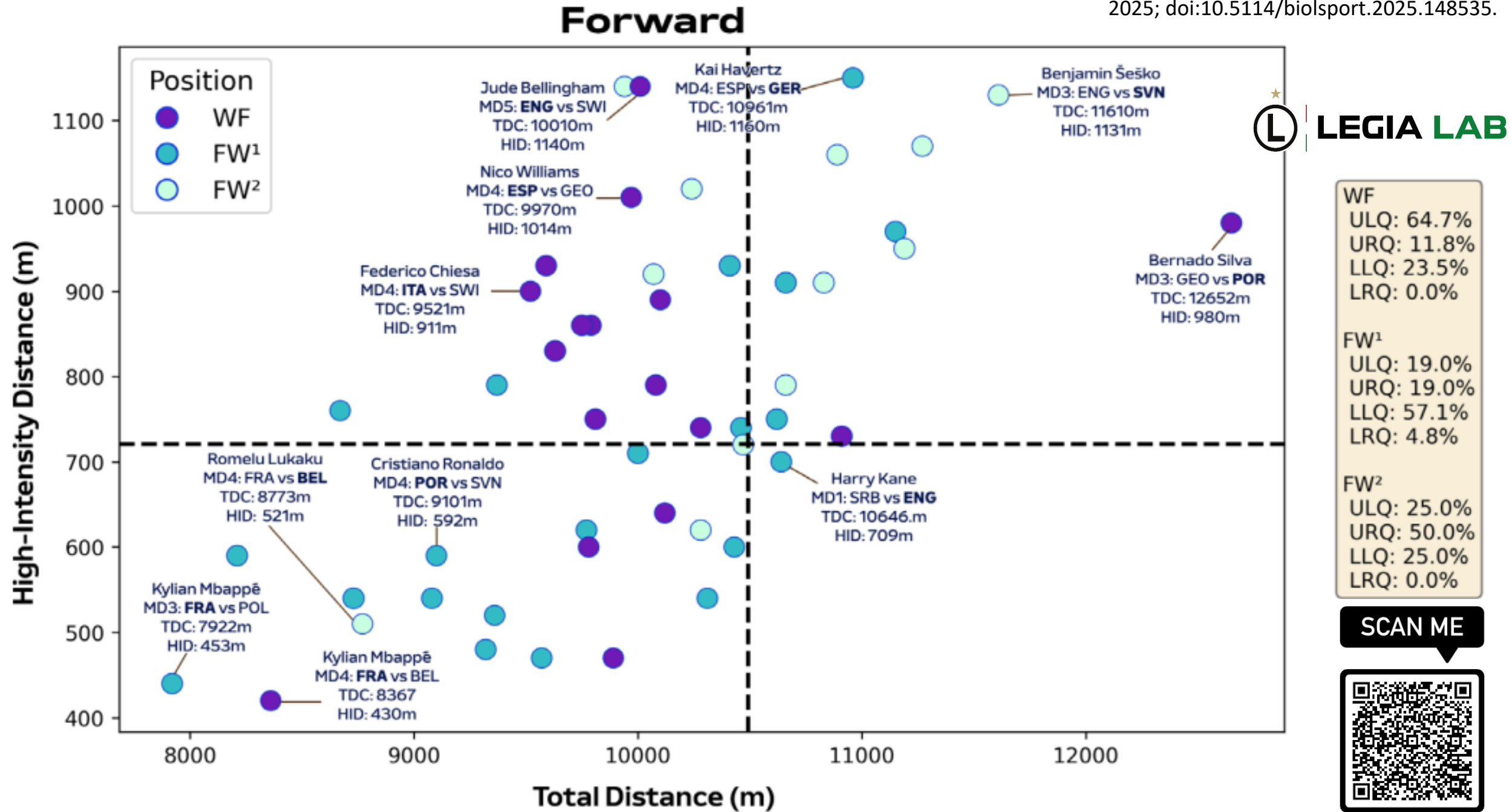
📊 **Analyzed Performance Metrics:** total distance covered, HSR, sprint distance, peak 1-minute distance, peak speed, technical and tactical actions (e.g. passes, crosses, dribbles, tackles, shots, penalty area entries).





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TABLE 1. Match events for each specialised positional role at UEFA Euro 2024.

Variables	CB ⁴	CB ⁰³	CB ⁶³	FB	WB	DM	DM ¹	CM	WM	AM	WF	FW ¹	FW ²	Basic Summary of <i>P</i> -values
<i>Distribution Events</i>														
Total Passes (No)	61.7±24.3	51.2±21	52.3±21.5	48.3±19.4	36.5±12.2	63.4±23.8	61.4±31.1	49.6±19.8	43.5±13.0	41.0±16.4	36.4±16.2	19.9±9.0	16.4±7.5	FW ^{1*} , FW ² <All [#] Except WB, WM, WF. DM, DM ¹ >FW ^{1*} , FW ^{2*} .WB [*] , WF [*] .
Passes Completed (%)	91.3±7.0	87.6±7.4	91.3±6.4	86.8±8.2	78.5±10.1	91.4±6.7	88.0±7.6	87.2±7.1	88.6±8.5	83.4±8.9	78.2±13.5	80.9±12.2	76.1±13.1	DM, CB ⁴ , CB ^{3c} >AM [*] , FW ^{1*} , FW ^{2*} . WB [*] , WF [*] . CB ^{3o} , CM>FW ^{2*} . WB [*] , WF [*] .
Total Pass Forward (No)	19.9±9.8	19.9±9.8	16.7±7.0	15.6±6.9	13.0±6.1	17.3±9.2	15.5±6.2	14.0±6.5	11.0±2.1	10.4±5.4	7.9±4.4	4.1±3.2	3.0±2.5	FW ^{1*} , FW ² <All [#] Except WM, WF. CB ⁴ , CB ^{3o} , CB ^{3c} >AM [*] ,CM [#] , WB [#] , WF [#] .
Pass Forward Completed (%)	79.5±13.5	77.1±15.0	81.3±12.7	75.6±16.5	64.7±18.7	80.7±16.2	77.16±13.8	73.0±15.5	77.2±15.4	68.2±21.2	62.71±22.9	62.2±31.5	56.9±39.2	FW ^{1*} , FW ² <CB ^{4*} , CB ^{3o} [#] , CB ^{3c} [*] , DM [*] .
Total Crosses (No)	0.2±0.6	0.7±0.9	0.1±0.3	2.2±2.2	2.7±2.9	1.1±2.3	0.5±0.9	1.4±1.5	4.2±3.9	3.6±3.6	3.8±2.4	1.0±1.6	1.0±1.2	WF, AM, WM>CB ^{4*} , CB ^{3o} [#] , CB ^{3c} [*] , DM [*] , DM ^{1*} , CM, [*] FW ^{1*} , FW ^{2*} .
<i>Ball Touch Events</i>														
Dribbling (No)	0.3±0.6	0.4±0.8	0.1±0.3	1.0±1.4	1.5±1.6	0.9±1.1	0.7±0.5	0.9±1.2	2.0±1.4	2.4±1.9	4.3±3.7	1.9±1.7	2.9±2.6	WF, FW ² , AM>CB ^{4*} , CB ^{3o} [*] , CB ^{3c} [*] , DM [*] , DM ^{1*} , CM [*] , FB [*] .
Total Ball Touches (No)	121.3±44.4	105.4±37.4	102.1±40.0	106.5±36.7	85.5±23.0	122.0±46.2	115.7±52.9	98.4±38.0	93.8±23.0	89.5±28.8	83.3±31.8	49.7±19.1	44.6±17.4	CB ⁴ , DM>FW ^{1*} , FW ^{2*} , WB [*] , WF [*] . CB ^{30*} , CB ^{3c} [*] , DM [*] , DM ^{1*} , FB [*] >FW ^{1*} , FW ^{2*} .
One Touch Play (No)	11.9±4.8	11.4±5.3	10.8±4.3	10.9±4.6	10.4±4.4	14.8±5.4	17.6±7.1	14.3±5.1	9.2±5.5	11.3±5.9	7.9±5.1	7.1±4.1	7.1±2.9	DM, DM ¹ >AM [#] , CB ⁴ [#] . CB ^{3o} [#] , CB ^{3c} [#] , FB [*] , FW ^{1*} , FW ^{2*} , WB [*] , WF [*] . CM>FB [#] , FW ^{1*} , FW ^{2*} .
<i>Offensive Events</i>														
Total Shots (No)	0.5±0.7	0.5±0.8	0.5±0.8	0.4±0.7	0.4±0.7	1.1±1.1	0.5±0.7	1.3±1.2	1.2±1.2	2.3±1.7	2.5±1.9	3.3±2.1	2.3±1.5	WF, AM, WM, FW ^{1*} , FW ² >CB ^{4*} , CB ^{3o} [*] , CB ^{3c} [*] , DM [*] , DM ^{1*} , CM [*] , FB [*] .
Shots on Target (No)	0.1±0.4	0.1±0.4	0.1±0.4	0.1±0.3	0.1±0.4	0.3±0.5	0.1±0.3	0.5±0.9	0.3±0.5	0.6±0.8	0.9±0.9	1.4±1.4	0.6±0.5	FW ¹ , WF>CB ^{4*} , CB ^{3o} [*] , CB ^{3c} [*] , DM [*] , DM ^{1*} ,FB [*] . WB. FW ¹ >FW ² [#] .
Solo Runs: Attacking Third (No)	0.8±1.2	1.1±1.5	0.3±0.5	1.6±1.5	1.4±1.2	1.4±1.5	1.3±1.4	1.8±1.9	1.2±1.2	2.1±1.9	2.2±1.8	1.1±0.8	1.8±2.3	CB ⁴ , CB ^{3c} <AM [*] , CM [#] , FB [*] , WF [#] . CB ^{3o} <AM [#] .
Solo Runs: Key Play Area (No)	0.6±1.3	0.6±1.0	0.1±0.5	0.9±1.2	0.6±0.8	1.4±1.7	1.3±1.6	1.5±1.8	1.3±1.5	2.0±1.9	2.6±1.6	1.2±1.7	1.4±2.2	CB ⁴ , CB ^{3c} , CB ^{3o} <AM [*] , WF [*] .
Solo Runs: Penalty Area (No)	0.0±0.1	0.1±0.2	0.0±0.0	0.3±0.6	0.3±0.7	0.2±0.5	0.0±0.0	0.4±0.6	1.0±0.9	0.9±1.0	1.9±1.3	0.7±1.4	0.7±0.8	WF>All [*] Except WM.
<i>Defensive Events</i>														
Ball recovery (No)	5.6±2.7	5.1±2.3	6.4±2.7	4.9±2.3	3.7±2.4	4.8±2.4	4.6±2.9	3.2±2.0	2.8±1.7	2.2±1.5	1.4±1.2	0.8±0.9	1.0±0.9	CB ⁴ , CB ^{3o} , CB ^{3c} >WB [*] , CM [*] , AM [*] , WF [*] , FW ^{1*} . FW ^{2*} .
Tackles (No)	1.1±1.1	1.4±1.1	1.0±0.9	1.3±1.4	1.5±1.4	1.8±1.4	1.8±1.5	1.8±1.5	1.7±0.8	1.2±1.3	0.8±1.0	0.1±0.4	0.7±1.2	FW ¹ <CB ^{3o} [*] , FB [#] , WB [#] , DM [*] , DM ¹ [#] , CM [*] [#] .
Aerial duel (No)	1.5±1.4	1.8±1.5	2.0±1.8	0.8±1.0	1.0±1.1	1.0±1.2	0.5±0.5	0.6±1.1	0.2±0.4	0.8±1.3	0.6±0.7	1.7±2.0	1.4±1.3	CB ⁴ , CB ^{3o} , CB ^{3c} >CM [*] , DM [*] , AM [*] , WF [#]

All values have been rounded up or down to one decimal place. *P*<0.05[#]; *P*<0.01^{*}

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Chen S, Zmijewski P, Bradley PS. Biol Sport. 2025;
doi:10.5114/biol sport.2025.148535.

📊 3. Key Findings and Conclusions

◆ Positional physical trends:

- **Central and defensive midfielders (CM, DM):** exhibit the highest total distance covered (~11.4–11.9 km).
- **Wing-backs and full-backs (WB, FB):** Recorded the highest HSR and sprint distances (~1200–1500 m).
- **Center-backs (CB):** Demonstrated the lowest total distance covered and overall locomotive intensity (~9.7–10.1 km, HSR < 800 m).
- **Attacking midfielders (AM):** intensity > total volume (the highest frequency of offensive actions).
- **Forwards (FW):** in two-striker systems (FW2) significantly greater physical intensity compared to those in a lone-forward role (FW1).

◆ Technical and Tactical Performance Indicators:

- Most passes: defensive midfielders (DM).
- Most dribblings and penalty area penetration: wingers (WF) and attacking midfielders (AM).
- Most tackles: center-backs (CB).

📌 Practical Implications

✓ Position-specific training programs should be designed to reflect:

- Differentiated volume and intensity based on specific positional requirements.
- Match-Critical Phases (e.g., peak 1-minute intensity intervals).
- Technical-Tactical Context and individual player tasks (e.g., overlapping WB vs static CB).

✦ These findings provide a framework for designing evidence-based, position-specific training protocols derived from elite-level match data.

Evolving running load demands and fixture congestion in the English Premier League: a decade of insights from 2015/2016 to 2024/2025

Tom Allen ^{1,2} Matt Taberner ³ Mikhail Zhilkin,¹ Matthew Green⁴

Research Objective

- Description of **physical load demands** and **match schedule density**
- Context: **English Premier League (EPL)** teams
- Ten-season period: **2015/2016 – 2024/2025**

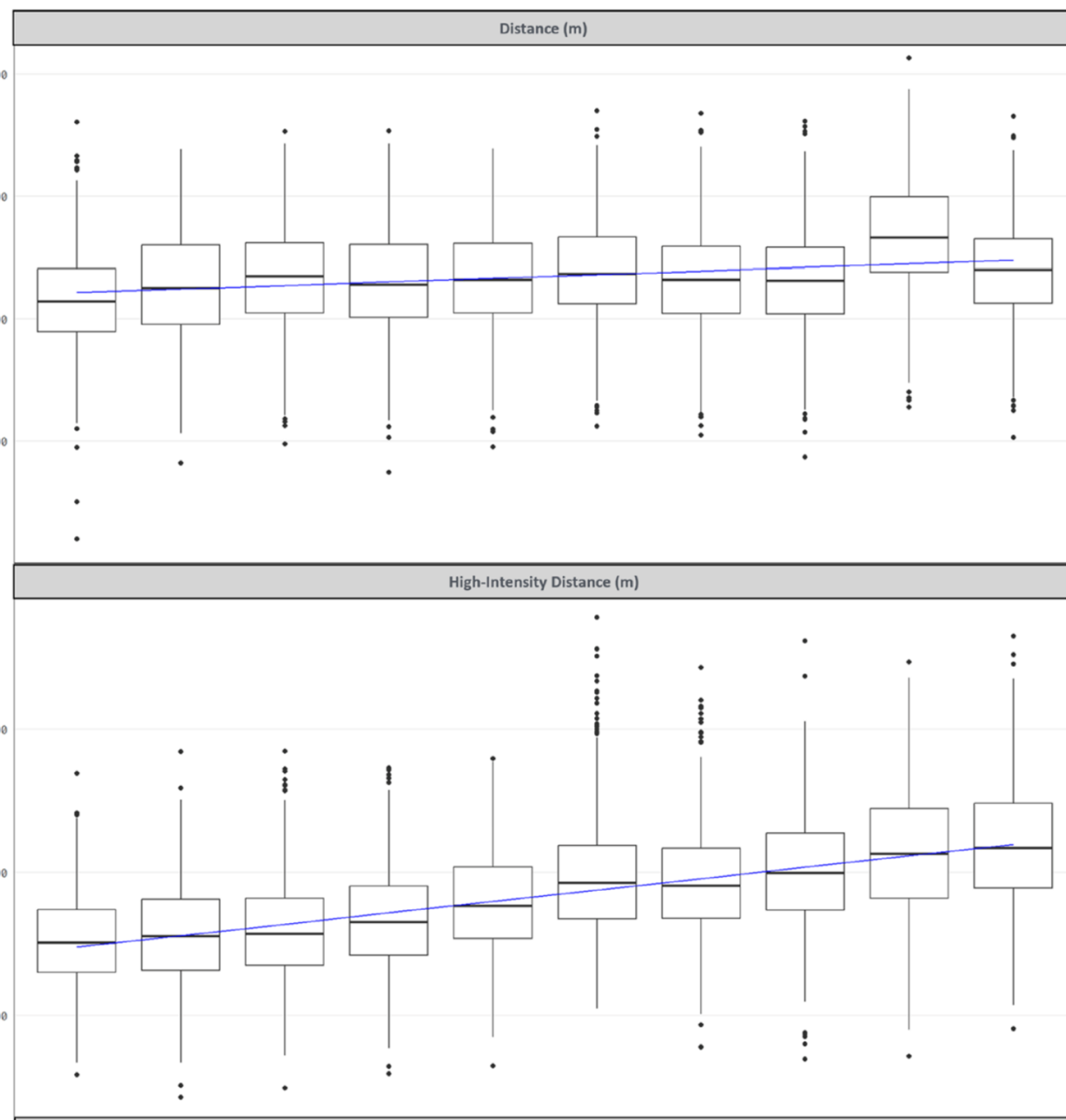
Methods

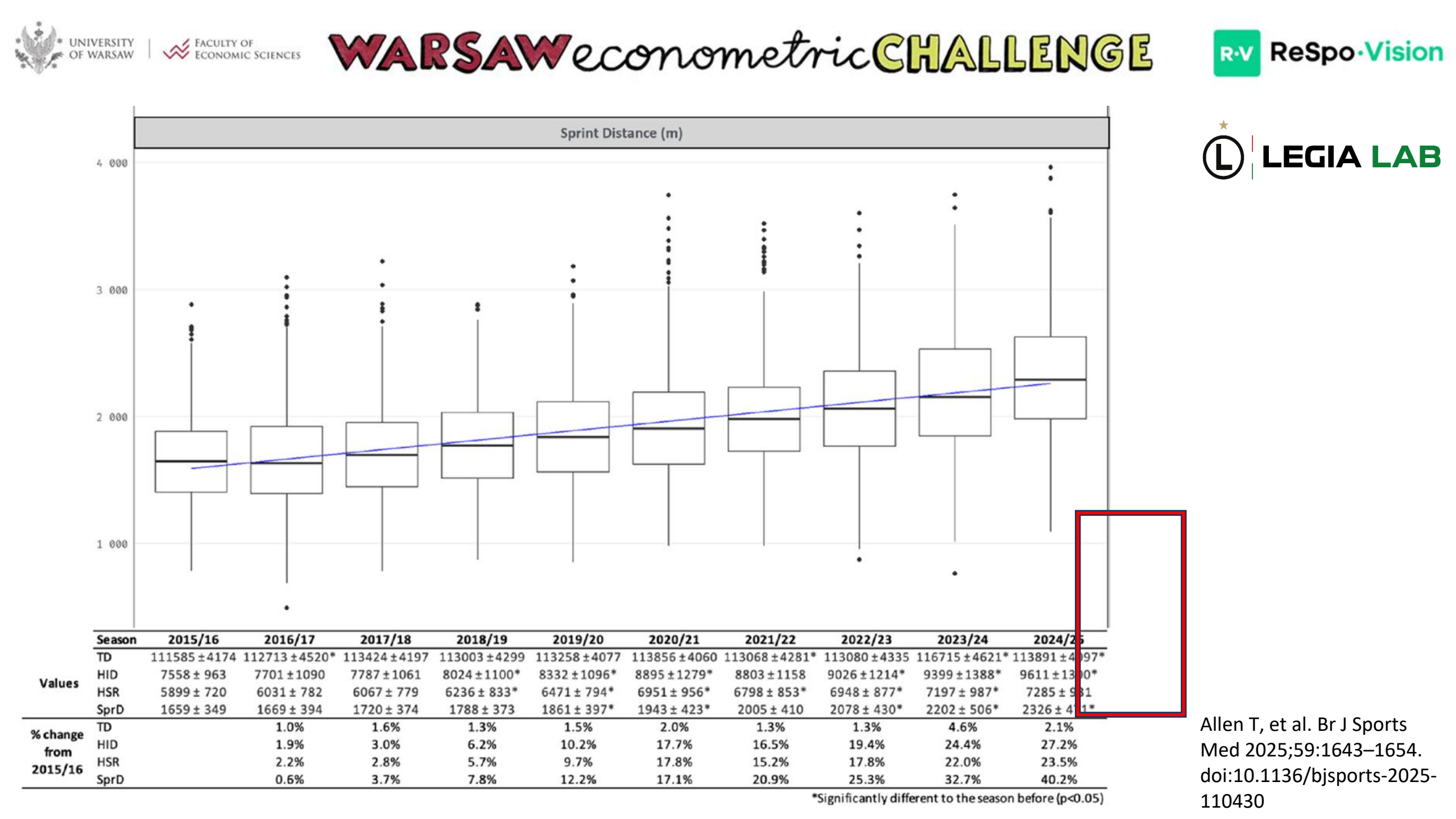
- **3750 matches**, over **10 seasons**
- Tracking Systems:
 - **TRACAB** (2015/16–2018/19)
 - **Second Spectrum** (2019/20–2024/25)
- Match Participation Data: **Wyscout**
 - Analysis of **total playing minutes** and **minutes played under conditions of fixture congestion**.

Current state of knowledge

- ↑ Rising trends in physical demands in EPL (2006–2018)
 -  Expansion of competition calendars by governing bodies:
 - **UEFA**
 - **FIFA**
- ⇒ Exacerbated **fixture congestion**.

Allen T, et al. Br J Sports Med 2025;51:1654. doi:10.1136/bjsports-2025-110





Congested Minutes: All Players, Top 6 Teams (T6)

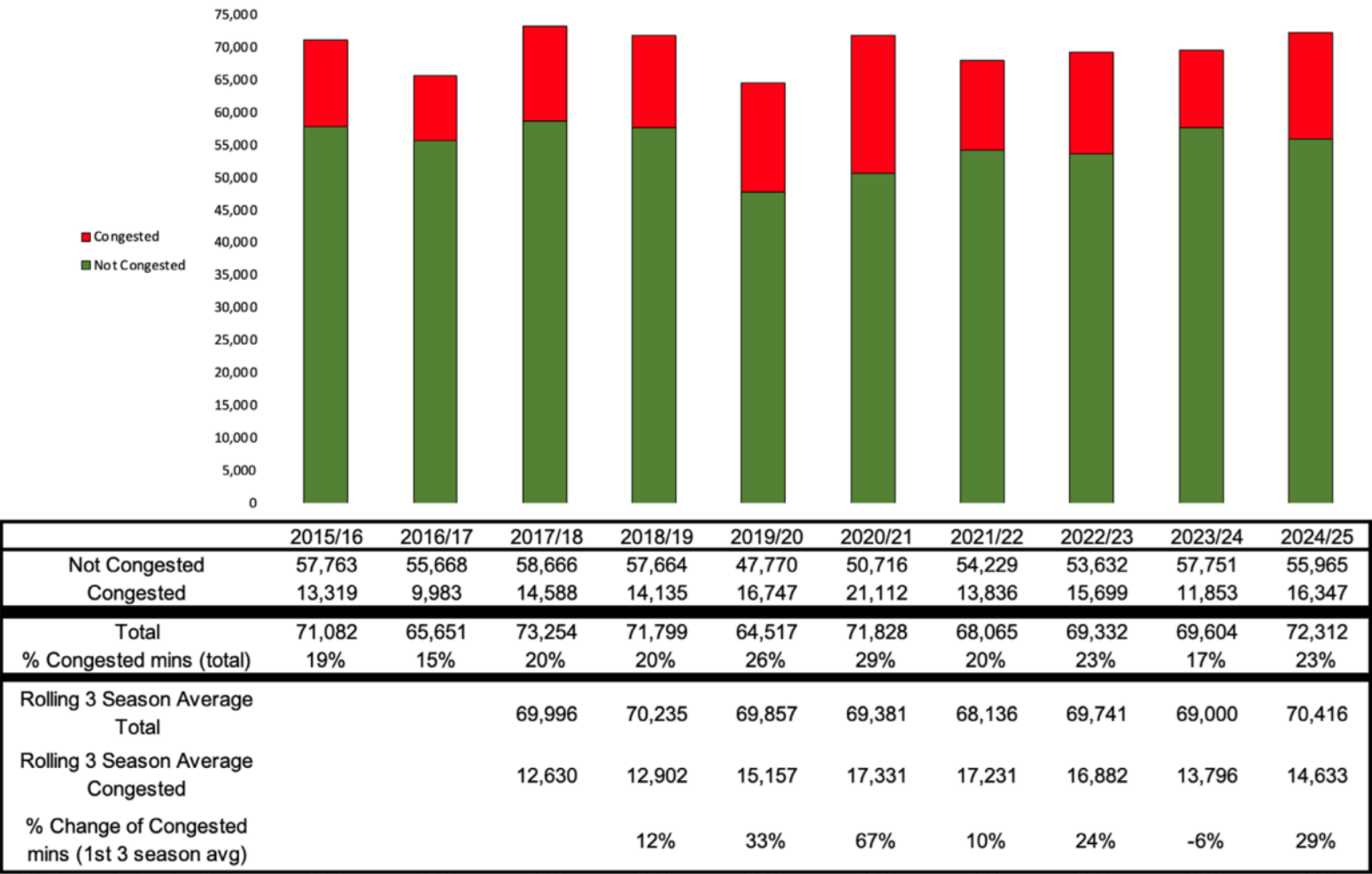


Figure 5 All players (minimum five appearances and >270 mins in EPL) and top six team players (T6; players from top six clubs) non-congested and congested (three games in 8 days) minutes per season in competitions. All players, all clubs— sum of all minutes/20. All players, top six teams— sum of all minutes/6. EPL, English Premier League

Allen T, et al. Br J Sports Med 2025;5:1654. doi:10.1136/bjsports-2025-110



📈 Increased Performance Demands (Past Decade)

- ↑ Physical load metrics: **TD, HID, HSR, SprD**
- ↑ Relative intensity metrics: **HID/min, HSR/min, SprD/min**
- ↑ **Total playing time**
- ↑ **Minutes played under congested scheduling (match congestion)**

🏆 Inter-team differences

- **Top 6 teams:**
 - higher **cumulative match loads**
 - greater volume of **minutes played under conditions of fixture congestion**
- **vs. the remaining teams**

🔍 Implications for Monitoring

- 🔍 **Continuous monitoring of athlete responses** is essential
- ⚠️ **Critical for health assessment**
- 🎯 **Priority focus:**
 - athletes with **the highest exposure**

Original research



Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: the UEFA Elite Club Injury Study from 2001/02 to 2021/22

Jan Ekstrand ,¹ Håkan Bengtsson ,¹ Markus Waldén ,¹ Michael Davison,²
Karim M Khan ,³ Martin Hägglund ¹

Research Objective

- Epidemiology and injury burden of hamstring strains in professional football: A 21-season study (2001/02–2021/22)
- Temporal trend analysis covering the last eight seasons (2014/15–2021/22)
- Injury Characteristics:
 - Injury location
 - Mechanism of injury
 - Recurrence rate



Methods

- **N = 3909 players, 54 teams, 20 European countries**
- **Period: 21 consecutive seasons (2001/02–2021/22)**
- Data Collection by Medical Staff:
 - Player exposure
 - Time-loss injuries
- Trend analysis:
 - **Poisson regression**
 - Generalized linear models (GLM)

Results: The Magnitude of the Problem

- **2636 hamstring injuries = 19% of all recorded injuries**
- Proportional increase:
 - from **12%** (initial season) → **24%** (final season)
- Increase in total time-loss days:
 - from **10%** → **20%**

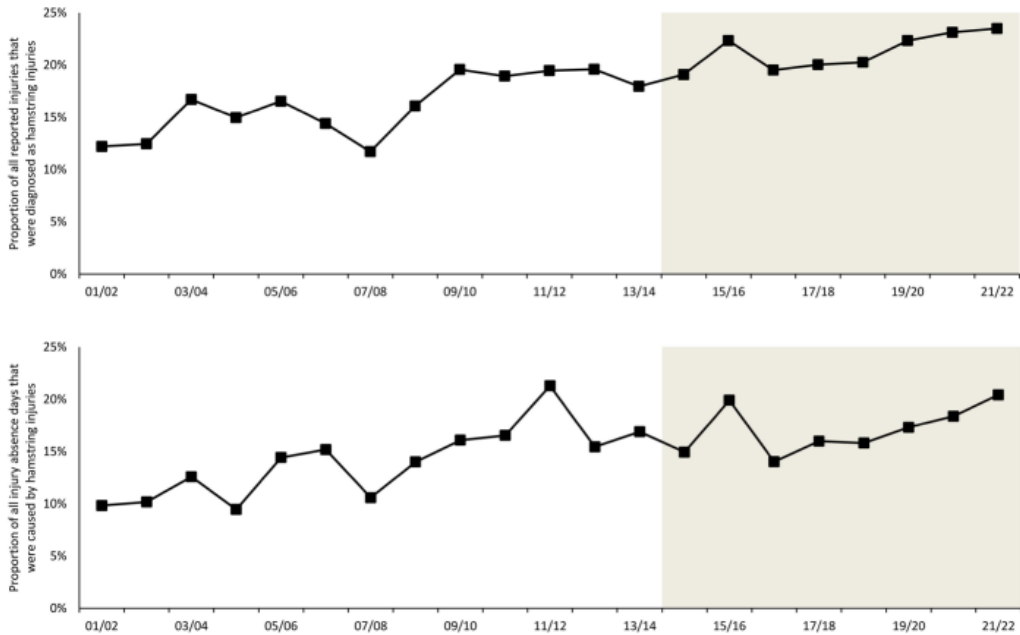


Figure 2 Development of the proportion of all reported injuries that were diagnosed as hamstring injuries and the proportion of all injury absence days caused by hamstring injuries over the study period. The shaded area represents the most recent eight-season period which has not been previously published.

Results – trends (2014/15–2021/22)

• Training:

- ↑ Injury incidence: **+6.7% per annum** (95% CI: 1.7–12.5%)
- ↑ Time-loss: **+9.0% per annum** (95% CI: 1.2–18.3%)

• Match:

- ↑ Injury incidence: **+3.9% per annum** (95% CI: 0.1–7.9%)
- ↑ Time-loss: : **+6.2% per annum** (trend, n.s.)

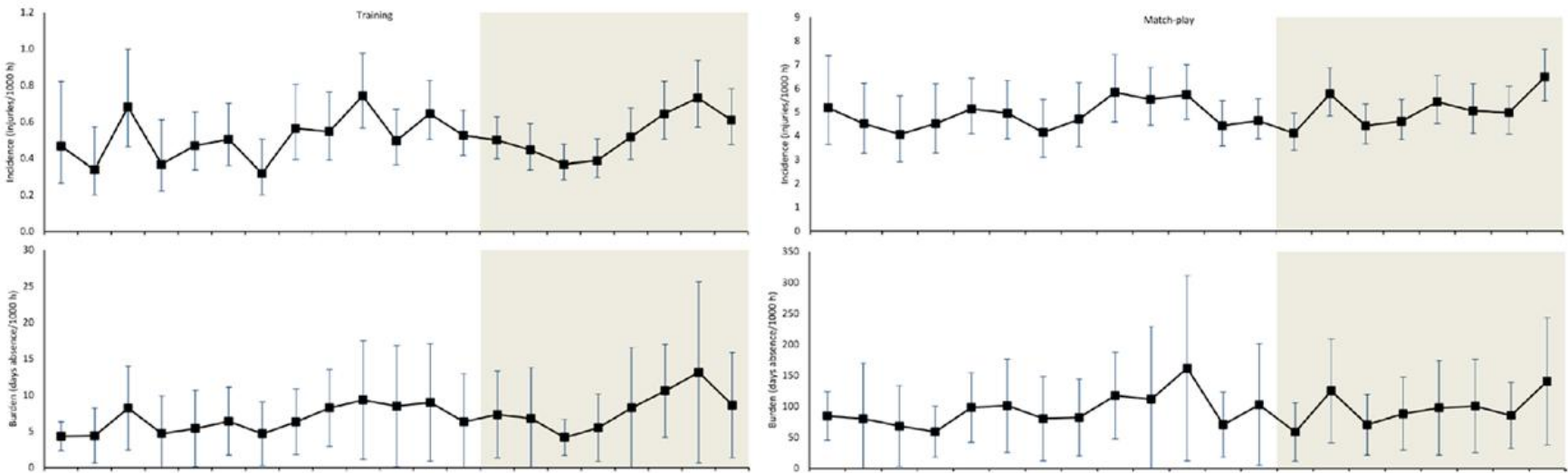


Figure 1 Development of hamstring injury incidence and injury burden over the study period. Injury incidence is defined as the number of injuries per 1000 hours of exposure, presented with 95% CI. Injury burden is defined as the number of absence days caused per 1000 hours of exposure and is presented as the mean of participating teams with SD. The shaded area represents the most recent eight-season period which has not been previously published.

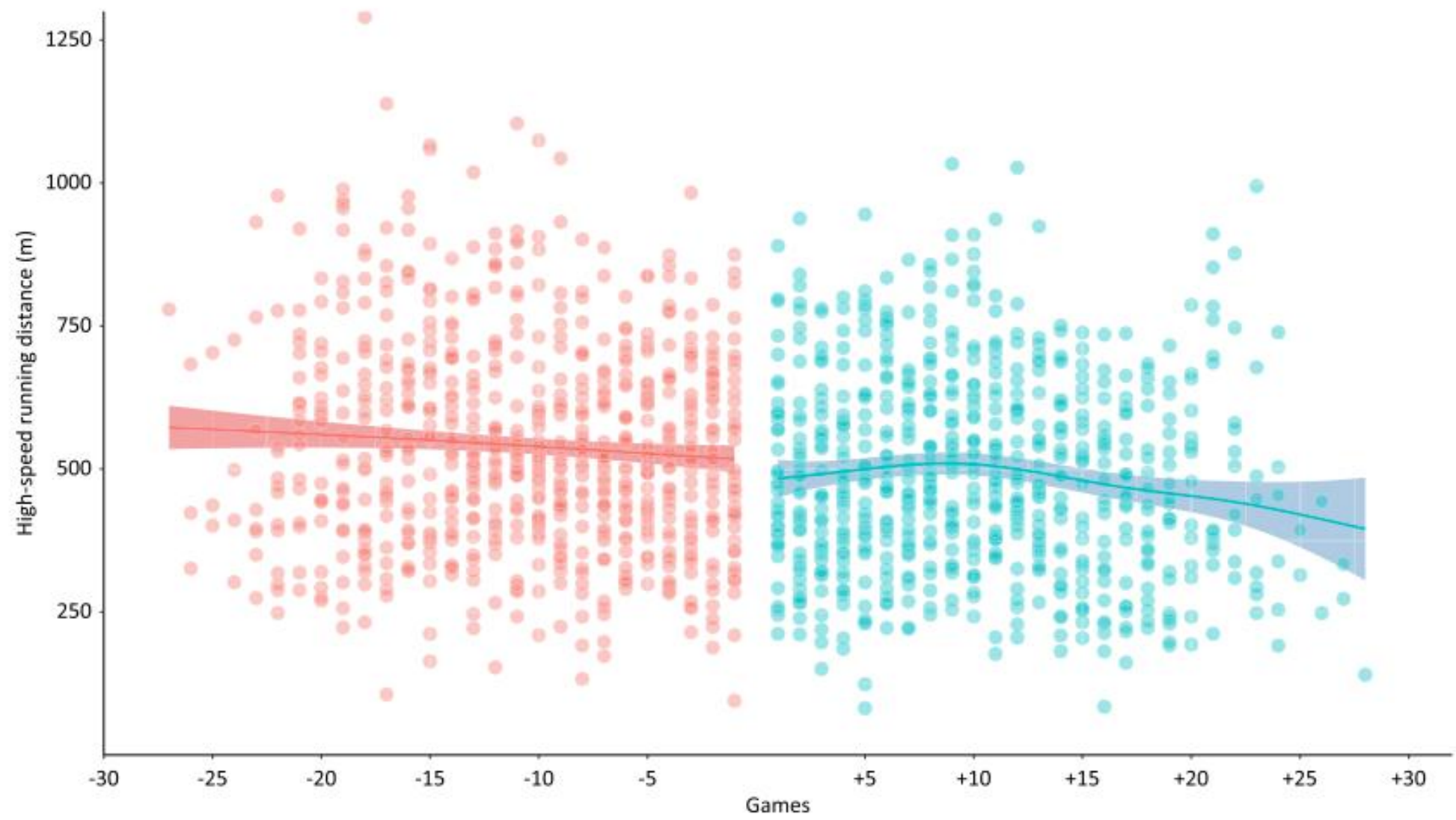


FIGURE 3 Individual data points for total high-speed running distance (meters covered at >20 km/h) in the game prior and subsequent to hamstring strain injury along with the lines of best fit and their 95% confidence intervals

Whiteley R et al. Scand J Med Sci Sports. 2022;32:1502–1509.

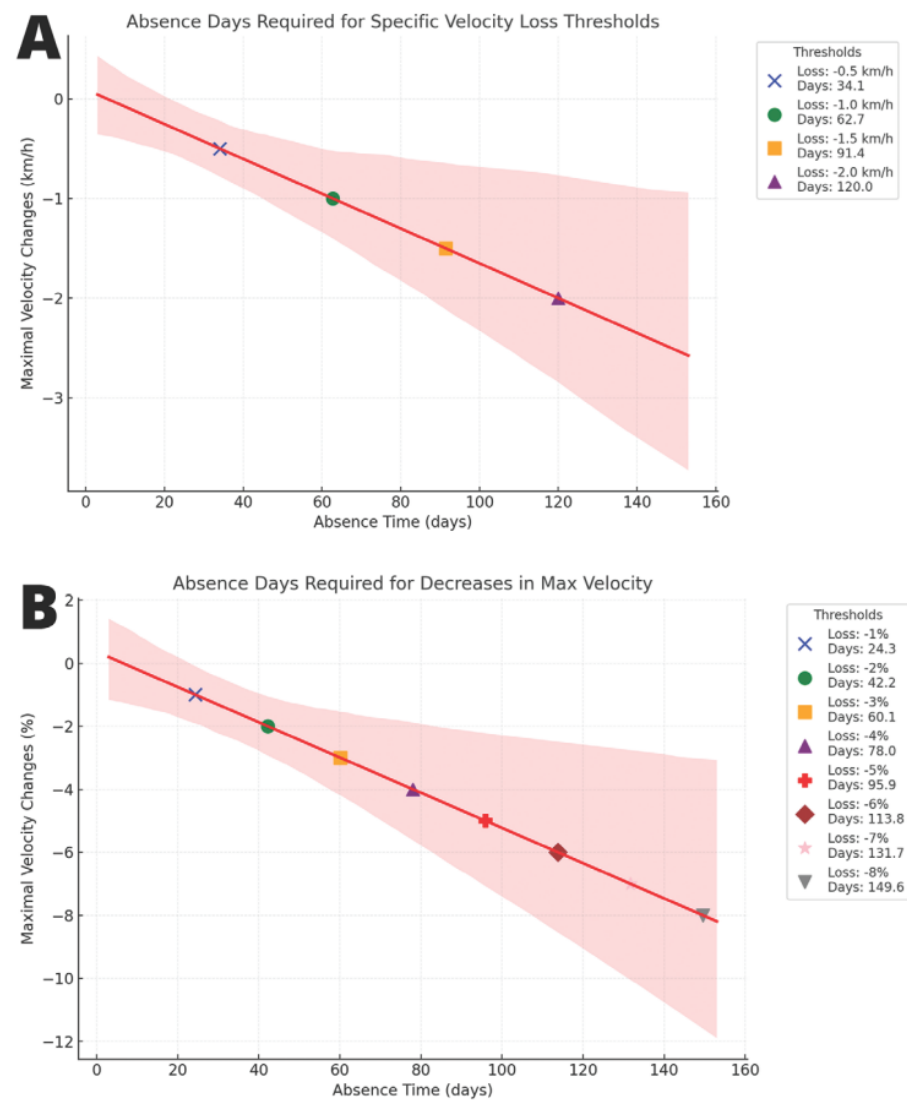


FIG. 3. Linear regression models illustrating the effects of absence time on maximal speed changes, with 95% confidence intervals. Plot A shows changes in absolute values (km/h), while plot B represents changes in percentages of pre-injury values.

Pecci J, et al. Biol Sport. 2025;42(4):275–286.

Possession in motion: a five-season analysis of running in-possession and out-of-possession with match outcomes in the English Premier League



AUTHORS: Tom Allen^{1,2}, Matt Taberner³, Mikhail Zhilkin¹, Damian Harper², Jill Alexander² | Corresponding author: Tom Allen

TABLE 4. Correlations ($r \pm 95\%$ confidence intervals) between possession and running load variables to points per game over five seasons in the English Premier League (EPL).

Possession and Running Load Variables	<i>r: correlation to PPG</i>	lower	upper	<i>p-value</i>
Possession	0.74*	0.64	0.83	< 0.001
TD TOOP/ TIP Ratio	0.65*	0.52	0.75	< 0.001
HSR TOOP/ TIP Ratio	0.62*	0.47	0.72	< 0.001
HID TOOP/ TIP Ratio	0.59*	0.44	0.70	< 0.001
BiP	0.51*	0.34	0.64	< 0.001
SprD TOOP/ TIP Ratio	0.50*	0.34	0.63	< 0.001
HSR TOOP	0.45*	0.28	0.59	< 0.001
HID TOOP	0.44*	0.27	0.59	< 0.001
SprD TOOP	0.41*	0.23	0.56	< 0.001
TD TOOP	0.34*	0.14	0.49	< 0.001
SprD	0.25*	0.05	0.42	0.010
HID	0.17	-0.02	0.36	0.085
HSR	0.13	-0.07	0.31	0.149
TD	0.04	-0.16	0.24	0.570
SprD TIP	-0.22*	-0.40	-0.03	0.023
HID TIP	-0.37*	-0.53	-0.19	< 0.001
HSR TIP	-0.41*	-0.57	-0.24	< 0.001
TD TIP	-0.57*	-0.68	-0.41	< 0.001

*p < 0.05; TD = team total distance, HSR = high-speed running, HID = high-intensity distance, SprD = sprint distance, TIP = team in-possession, TOOP = team out-of-possession, BiP = Ball in Play

Allen T, et al. Biol Sport. 2026;43:899–909.

Challenge Statements

Analysis 🔍📊

- **Load Quantification**

Robust metrics (TD, HSR, Sprint) normalized for context

- **Context Modeling**

Explain performance via match status, tactics, opponent

- **Fixture Congestion**

Assess impact of short recovery on performance

- **Player Profiling**

Individualized models of running output

- **Peak Demands**

Identify worst-case scenarios

Application ⚙️📋

- **Injury Risk Indicators**

Link load and congestion with injury risk

- **Data Integration**

Combine tracking + event data

- **Actionable KPIs**

Translate metrics into coaching insights

- **Visualization**

Clear dashboards for decision-making

- **Decision Support**

Models for rotation and load management



Thank you for your attention



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ECONOMIC SCIENCES

WARSAW *econometric* **CHALLENGE**

R·V ReSpo·Vision

R·V ReSpo·Vision

FOOTBALL TRACKING DATA

by Radosław Marcinkowski

OUR CREDENTIALS

**We've worked with absolutely
the best in the world**

Fully remote

Optical Tracking System: Players, Ball & Referees

Selected clients

Teams



Federations

FIFA



Concacaf

Data Companies



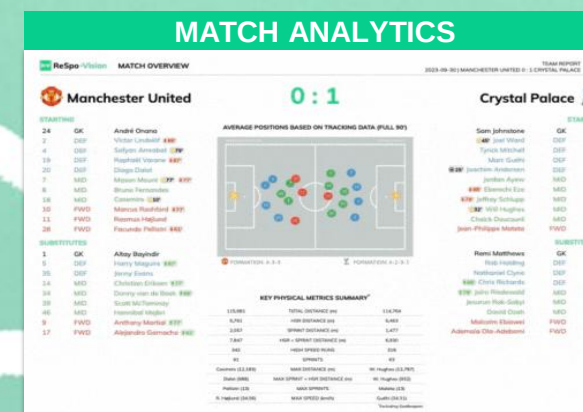
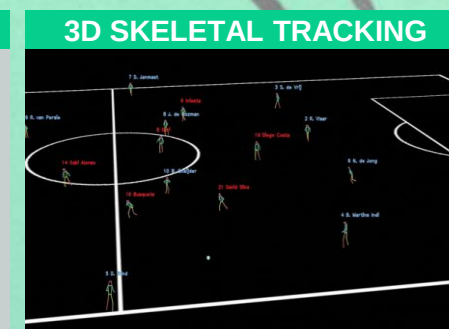
TV Broadcast



Wide-angle or pano



No wearables or additional hardware



Kognia



Data Analytics in Football

Cutting-Edge technology in football stats

- Sports data is ever-evolving: from simple manual stats to very accurate automated optical tracking
- Today tracking data is **used by the best to optimize sporting and financial decisions**
 - Performance analytics
 - Opponent analytics
 - Scouting
 - Injury Prevention

General stats



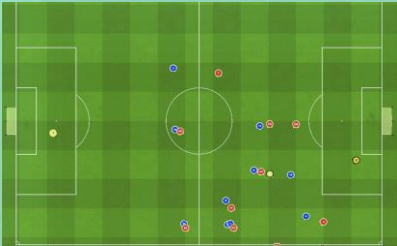
Event data



Telestration

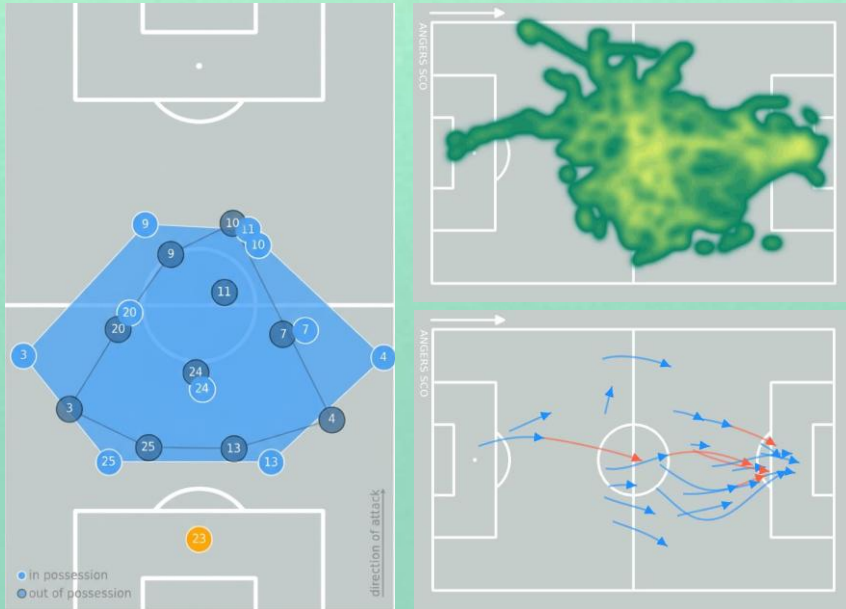


2D & 3D tracking



Football Tracking Data

Tracking is a natural evolution of event data and provides much deeper insights



- Tracking data provides information on the exact **2D coordinates (x,y) of each player and the ball** up to 50 times per second or even up to 40+ body parts of players with detailed 3D tracking capabilities.
- Tracking data **solve problem of the missing context** of the events data.
 - Where players perform their runs?
 - Are you available forward passes or lateral?
 - When our opponents are playing more intensively?
- It is crucial to know not only about the number of passes made or distance covered but **HOW, WHEN** and **WHERE** each event/run/pass had happened.



○ **Save:**

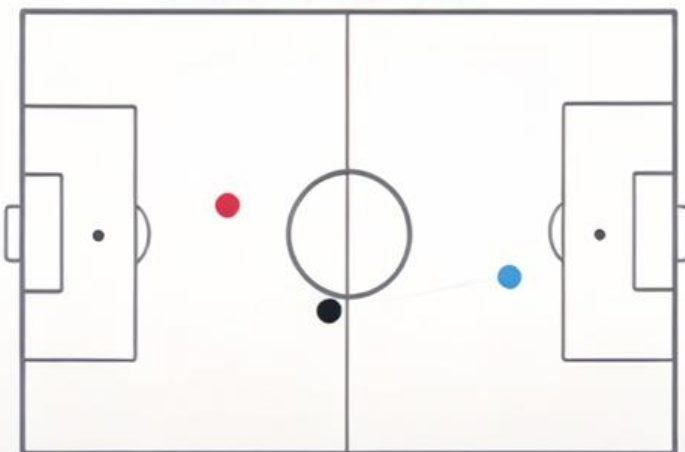
- Time ⌚
- Money 💰
- Resources 🧑

EVENT DATA VS TRACKING DATA.

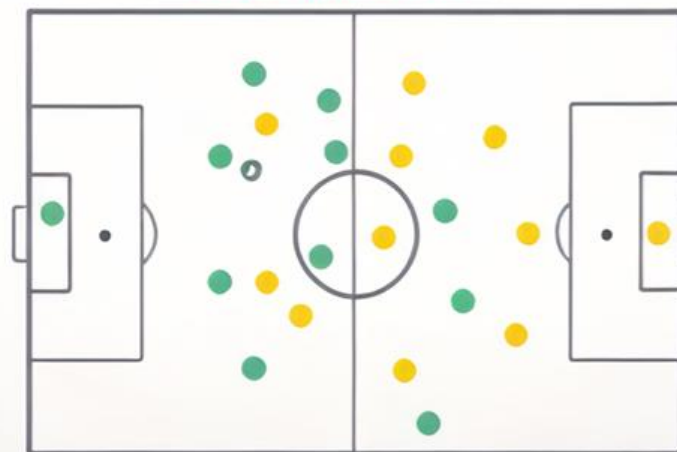
High Availability

Low Availability

BALL EVENT DATA



TRACKING DATA



Low Granularity

High Granularity



RESPO.VISION: OUR TECH IS ABSOLUTELY UNIQUE

First ever single camera 3D tracking system

1



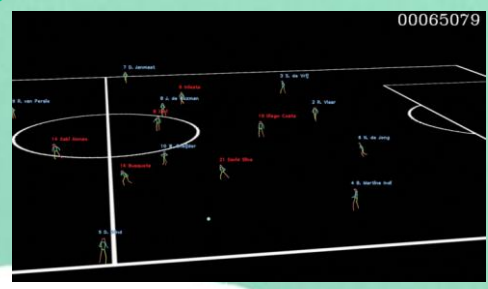
SINGLE CAMERA
FEED PROCESSING

2



PLAYER &
OBJECT
DETECTION

3



BODY PARTS &
POSE DETECTION



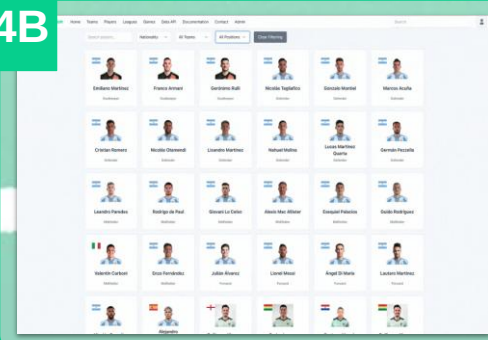
Quality and reliability certified by
FIFA Quality Programme
for EPTS
(world program & tactical feeds)

4A



ANALYTICS: REPORTS

4B



00010523

PUN 0-0 MOT 06:58

CANAL+

Kewin Komar 1

Piotr Mroziński 77

Dawid Abramowicz 33

Filip Wojcik 17

Roman Yakub

Samuel Mraz 94

Bartosz Germani 63

Jani Atanasov 6

Jakub Labojko 21

Dawid Szymonowicz 4

Bradly van Hooven 19

Kaan Gullakaner 11

Jakub Serafin 14

Puszczo Niepolomice

Mistrz Lublin

MAKING COMPLEX DATA UNDERSTANDABLE

PHYSICAL METRICS

Which player is the fastest/most explosive?

Who runs the most/least?

Does the player get exhausted in later phases of the match?

FORMATIONS' SHAPE & DISTANCES

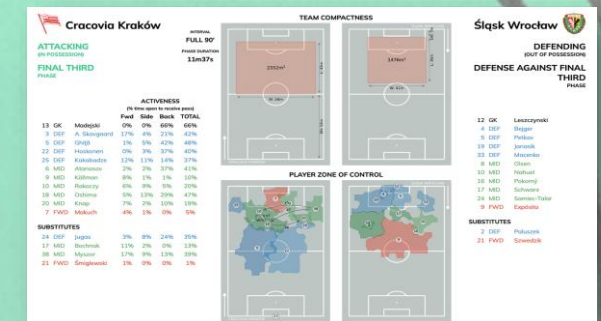
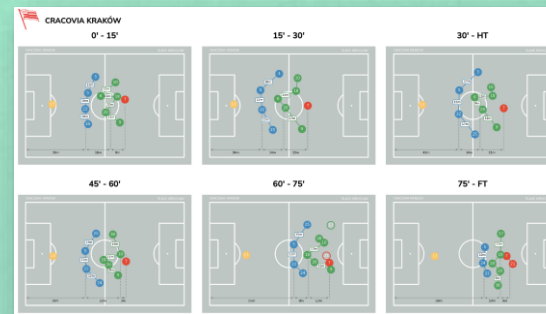
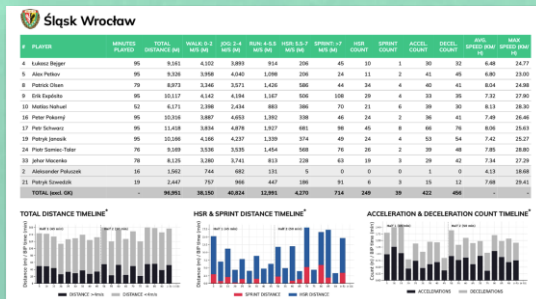
How does the team formation change over time?

Which players are positioned furthest apart, leaving space in between?

PLAYERS' ZONES OF CONTROL

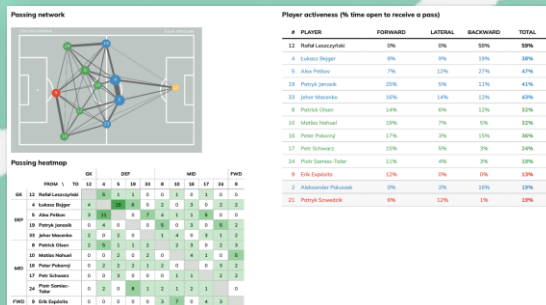
In which zones do we leave open spaces?

Where does the opponent play the ball in different phases of the game?



PASS AVAILABILITY

Which players are the most active - most often opening a passing lane to a teammate with the ball?

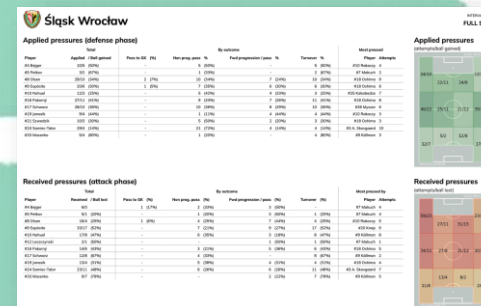


PRESSING ANALYSIS

How frequently do players press?

How do opposing players react to pressing?

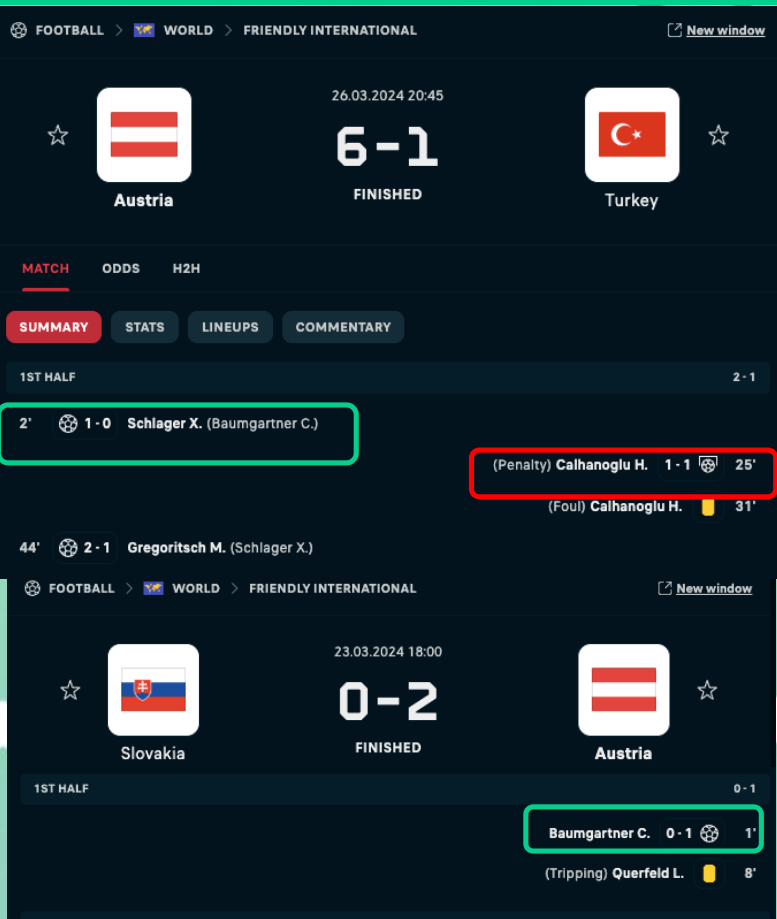
Do they pass to the goalkeeper or play the ball forward?



PRACTICAL EXAMPLES

Austrian national team intensity starts of the match

- Austria had a clear tendency to start the games intensively with a lot of HI-runs. This helped them to start each of the last 4 games before the EURO with a fast opener.
- However, they repeatedly paid for it in the 25'-35'th minute, when they struggled after an intensive start.



PRACTICAL EXAMPLES

Austrian national team intensity starts of the match

- This knowledge didn't help Poland to prevent this. They were not able to stop the fast and quick Austrian attackers, which led to a quick 0:1 goal.
- Luckily for them, they were able to take advantage later when Austria had their "classic fatigue crisis" around 25'-35'th minute and came back into the game.

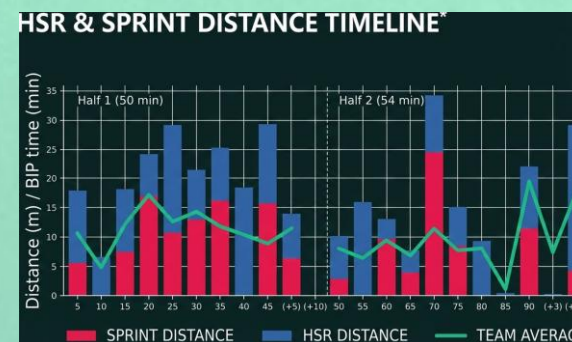
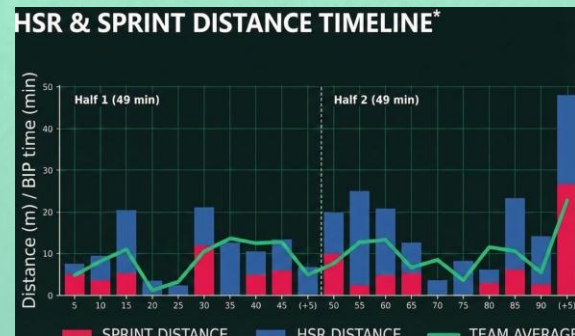
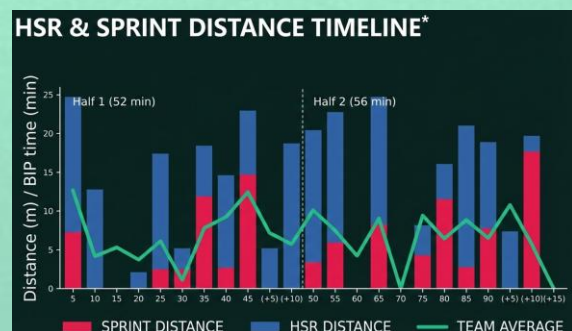




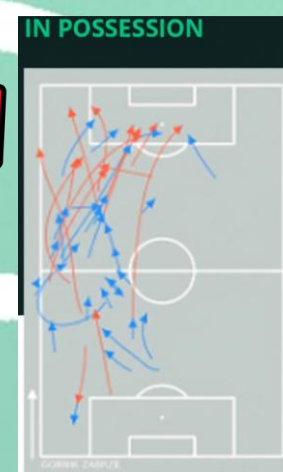
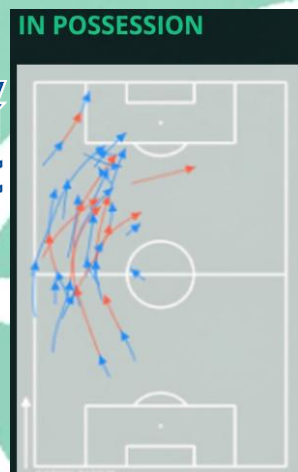
Tracking data

PRACTICAL EXAMPLES

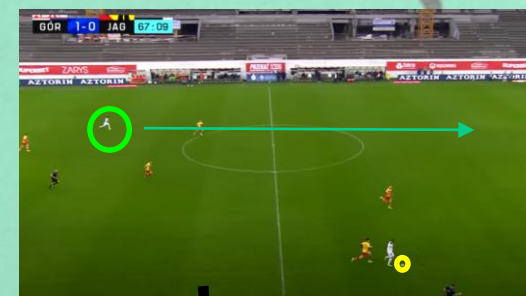
- High defensive line of opponents was kind of perfect for Ennali, who was able to engage in a lot of long races with defenders, leading to the goal against Jagiellonia.
- In those games he averaged 1,400 m distance in HI intensity.
- His speed is his superpower, as he reached a top speed of 34.7 km/h against Ruch Chorzów.



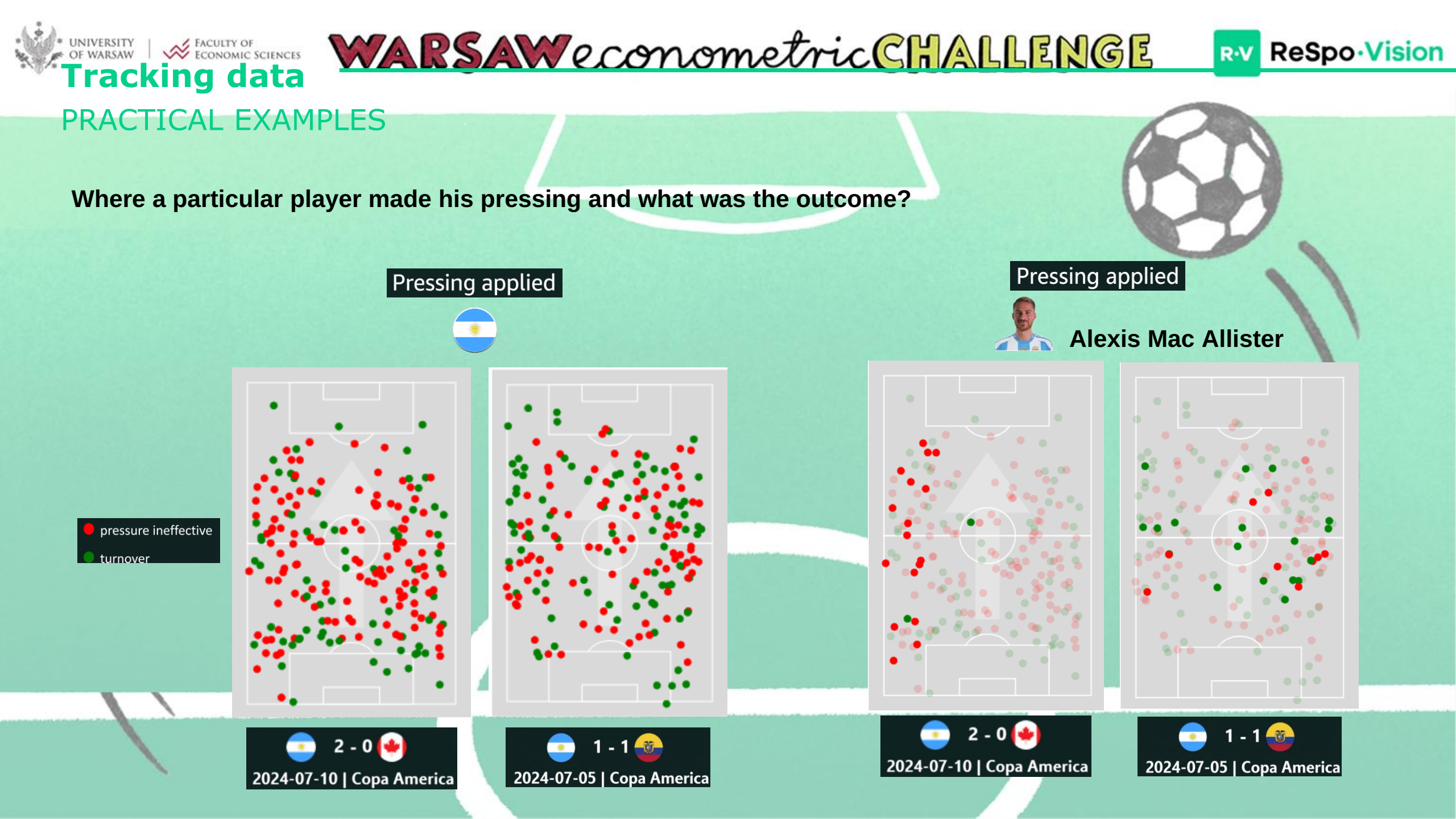
L. ENNALI running in High Intensity Maps & Timelines



11. Lawrence Ennali



68' Lawrence Ennali





GOOD LUCK



**COMPETITION STRATEGY
RECOMMENDATION
by Marcin Chlebus**

A few words about the competition strategy (1/5)

Team Tasks



Data Preparation

Preparing data for analysis and implementation.



MVP Solution

Developing the minimum viable product for testing.



Extra Solution

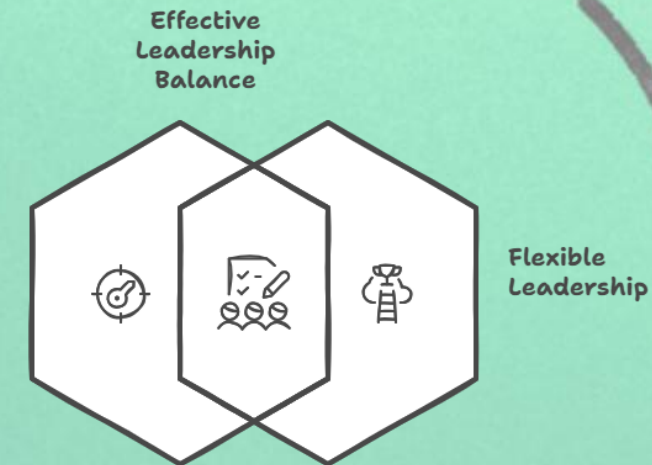
Creating additional features or enhancements.



Paper Writing

Documenting findings and results in a paper.

Effective Task Leadership



A few words about the competition strategy (2/5)



Data Preparation

1. **Identify Relevant Data Sources:** Start working on data from the very beginning. Work with transferable formats.
2. **Data Cleaning and Preprocessing:** Ensure that the data is clean and formatted correctly. This includes handling missing values, removing duplicates, and normalizing data types. Utilize tools like Pandas, dplyr or Excel for efficient data manipulation.
3. **Data Exploration:** Conduct exploratory data analysis (EDA) to understand the data's characteristics. Visualizations can help identify trends, patterns, and anomalies that may inform your solution.
4. **Documentation:** Keep information about main decisions about data sources, preprocessing steps, and any assumptions made during the data preparation phase. This will be invaluable for both the MVP and paper writing stages.

A few words about the competition strategy (3/5)



MVP Solution

1. **Define the Core Problem:** Clearly articulate the aims of the research, hypothesis, goals. This will guide the development of your Minimum Viable Product (MVP).
2. **Focus on Essential Solution:** Identify the key approaches/models that will demonstrate the viability of your solution.
3. **Iterative Development:** Use an agile approach to develop your MVP. Start with a basic version and iteratively extend the approach.
4. **Deliveries check-list:** Address all main research questions, iteratively consider the additional ones.

A few words about the competition strategy (4/5)



Extra Solution

1. **Identify Additional Features:** Once the MVP is established (but not developed), brainstorm additional features that could enhance the user experience or provide more value.
2. **Prioritize Based on Impact:** Evaluate which extra features would have the most significant impact and are feasible within the hackathon timeframe.
3. **Allocate Team Roles:** Assign team member(s) to work on extra features.

A few words about the competition strategy (5/5)



Paper Writing

1. **Outline the Structure:** Start with a clear outline that includes sections such as Introduction, Data, Methodology, Results, and Conclusion. This will provide a roadmap for your writing.
2. **Collaborative Writing:** Utilize collaborative tools like Google Docs to allow all team members to contribute to the paper. Assign sections based on individual strengths to streamline the writing process.
3. **Focus on Clarity and Conciseness:** Write clearly and concisely. Avoid jargon unless necessary, and ensure that complex ideas are explained in an understandable manner.
4. **Cite Sources and Acknowledge Contributions:** Properly cite any data sources, frameworks, or tools used in your project.



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WARSAW *econometric* **CHALLENGE**

R·V ReSpo·Vision

DESCRIPTION OF THE CONTEST PROBLEM AND PRESENTATION OF THE DATA

by Piotr Wójcik



Will you score a goal?

**Case study – predicting goalscoring probability
(UEFA U21 2025)**

Task description - Main Aim (1/4)



Coaches trust intuition. It works - but:

- ignores subtle performance patterns,
- misses interactions & context,
- is hard to compare objectively.

A **data-driven** approach enables a **more objective**, systematic and reproducible evaluation of player performance.

Aim: 🏆 *Build a predictive model to support coaches in better assessing a player's likelihood of scoring a goal over the remaining duration of the game*



Task description - What Data Do We Have? (2/4)



Tournament: UEFA U21 EURO 2025 (Slovakia)

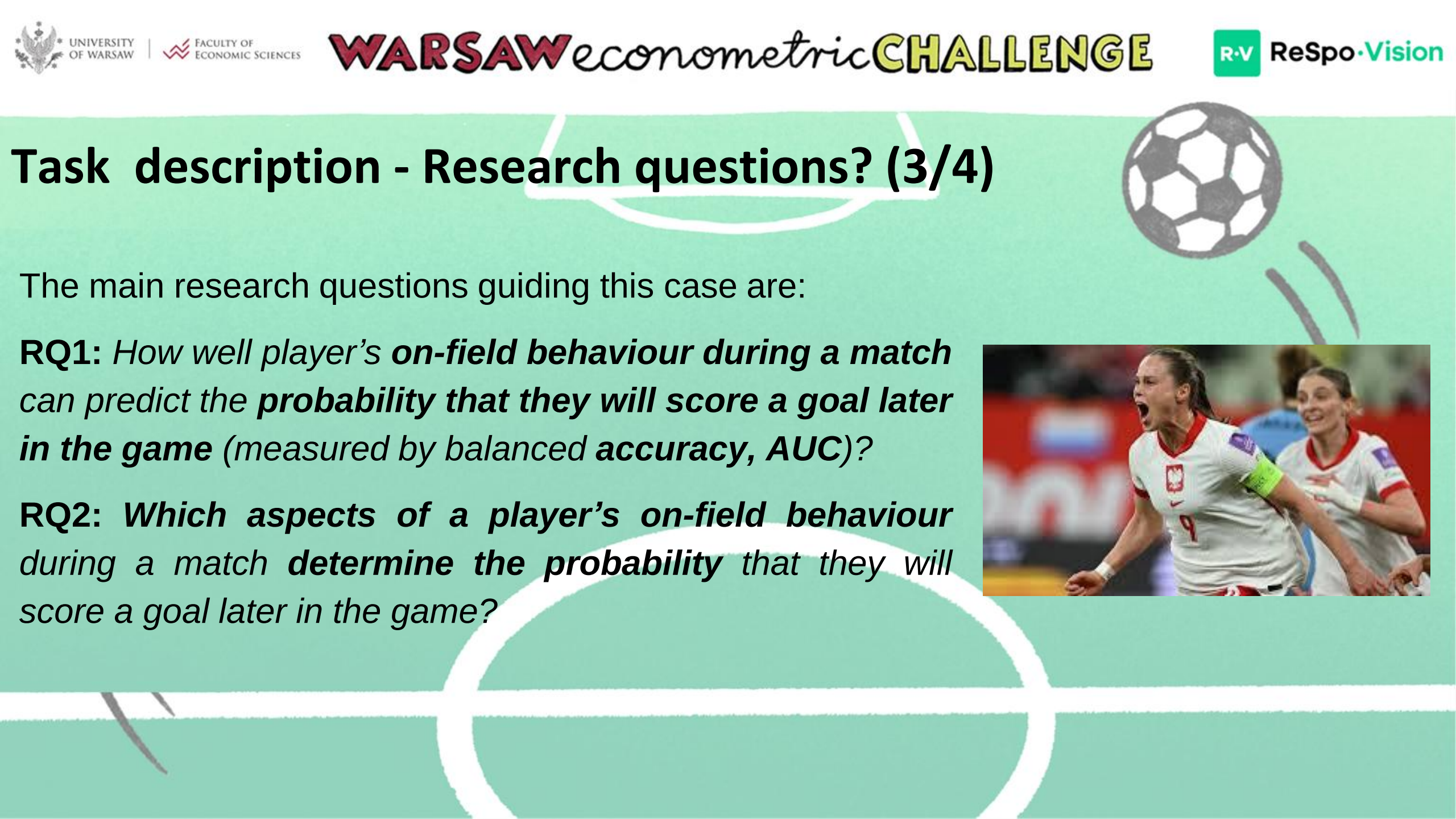
Match broken into: 15-minute intervals

Features:

- rolling **last_15** and **cumulative** features
- **supplementary datasets that include event-level data**
(shots, runs, passes and actions under pressure)

Target variable: SCORED_AFTER (goal 1 = yes / 0 = no)





Task description - Research questions? (3/4)



The main research questions guiding this case are:

RQ1: *How well player's **on-field behaviour** during a match can predict the **probability** that they will score a goal later in the game (measured by balanced **accuracy**, **AUC**)?*

RQ2: *Which aspects of a player's **on-field behaviour** during a match **determine** the **probability** that they will score a goal later in the game?*



Task description - Research questions cont? (4/4)



To gain deeper insights, several more detailed research questions can be formulated and addressed:

RQ3: Is information about **sprints and shots alone sufficient** to accurately predict the probability of a player scoring a goal?

RQ4: Does **incorporating passing data or information about player behaviour under defensive pressure** improve the predictive performance of goal-scoring models?

RQ5: Does **short-term performance (recent 15-minute activity) or cumulative performance up to a given point** in the match have stronger influence on the probability of scoring a goal?

RQ6: Does an **increase in short-term performance intensity relative to a player's overall performance level** increase the probability of scoring a goal?

RQ7: Do **external factors that are not directly controlled by the player** influence the probability of the **player scoring** a goal, and if so, which of them matter most?

Data description

There are **five** datasets representing various aspects of players' activity.

The **main dataset** for the contest:

- **players_quarters_final.csv** (3 486 rows) – player activity snapshots at match quarter-hour checkpoints



Source and supplementary datasets:

- **player_appearance_pass.csv** (29 795 rows) – individual pass events
- **player_appearance_behaviour_under_pressure.csv** (12 185 rows) – passes made while being pressed
- **player_appearance_run.csv** (35 133 rows) – high-speed run / sprint events
- **player_appearance_shot_limited.csv** (780 rows) – individual shot events

Data description – main dataset for the contest



- covers 31 matches and 869 unique player appearances
- each row represents **one player at one time checkpoint**, capturing their activity both in the **preceding 15-minute window** and **cumulatively from the start of the match** up to that point
- the binary **target variable scored_after** indicates whether the player scored a goal at any point **after the checkpoint** in the same match, excluding own goals
- constructed by aggregating event-level data from **selected source tables** (*player_appearance_shot_limited.csv* and *player_appearance_run.csv*)

Data description – main dataset for the contest



Observation unit and time structure

- checkpoints defined within match periods, with **minutes counted locally** from the start of each period
- up to **seven checkpoints per match**: minute 15, 30, and 45 of the first half (H1_15, H1_30, H1_45), minute 15 and 30 of the second half (H2_15, H2_30), and – for matches that reached extra time – minute 45 of the second half (H2_45) and minute 15 of the first extra time period (ET1_15)
- only players who were **actively on the pitch at the moment of the checkpoint** are included, identified through the continuous *minute_in* and *minute_out* columns

Data description – main dataset for the contest



Feature groups

- variables with the prefix ***last15_*** describe the player's activity in the 15-minute window **immediately preceding the checkpoint**
- variables with the prefix ***cumul_*** describe their accumulated activity **from the start of the match**
- both groups cover the same two domains: **physical exertion** (derived from *player_appearance_run.csv*) and **shooting** (derived from *player_appearance_shot_limited.csv*)
- Goals scored prior to the checkpoint were **excluded from the shot features** to prevent information leakage into the target variable

Data description – main dataset for the contest



Target variable

- ***scored_after*** is a binary indicator equal to 1 if the player scored at least one non-own goal **after the checkpoint** in that match, and 0 otherwise
- the target is heavily imbalanced: 203 positive cases out of 3,486 observations (approximately 5.8%), reflecting the natural **rarity of goalscoring** events in football

Data description – source and supplementary data



player_appearance_run (35 133 rows)

- shows how much **physical effort** each player puts in, by recording every high-intensity run they make
- the largest event table in the dataset — it reflects just how frequently players make explosive movements during a match
- each row represents a **single high-intensity run**, classified by *run_type* into two categories: *hsr* (high-speed running, intense but sub-maximal effort) and *sprint* (maximal or near-maximal speed)
- this table was used to create **some features** included in the main dataset for the contest, but they can be potentially **further extended**

Data description – source and supplementary data



player_appearance_shot_limited.csv (780 rows)

- logs **every shot attempt** across all matches in the dataset, with one row per shot event
- where this table really stands out is in the **detail it captures** about **how each shot was taken** – part of the body (right foot, left foot, head, or other), technique (volleys, lobs, and overhead kicks), etc. - giving a **full picture** of the tactical context in which the shot arose
- this table was used to create **some features** included in the main dataset for the contest, but they can be potentially **further extended**
- **outcome** of the shot **excluded** from the table provided for the contest

Data description – source and supplementary data



player_appearance_pass.csv (29 795 rows)

- the most granular passing log in the dataset, with **one row per pass attempt** made by a player during a match
- it captures **every time** a player did **play the ball to a teammate** or tried to
- there is a field that links to that **receiver's appearance** record (which can be null) and accuracy that identifies if the pass actually reached its intended target
- this table **was NOT used** to create features included in the main dataset for the contest

Data description – source and supplementary data



player_appearance_behaviour_under_pressure.csv (12 185 rows)

- this table zooms in on a specific and tactically rich subset: moments when a **player receives and attempts to play** the ball while **being actively pressed** by an opponent
- it reflects how frequently pressing occurs in these matches and how players react in difficult, high-pressure situations
- one can identify **who** applied the pressure and **what happened** afterwards
- this table **was NOT used** to create features included in the main dataset for the contest



Files on google drive

 data

 WEC2026_data_description.pdf 

 WEC2026_Problem_description.pdf 





WARSAW econometric CHALLENGE

 ReSpo·Vision

WEC 2026 problem description

Will you score a goal?

Predicting goal-scoring probability in football using performance match data

In modern professional sport and football in particular, real-time and post-match data analysis has become a standard analytical support tool used by most coaching staffs worldwide. One of the most important application areas of such analysis is the assessment of **player performance during a match**. Experienced coaches tend to rely on their intuition to identify which players are performing well and which are beginning to decline. While this intuition is often valuable, it is inevitably based on heuristic judgments that may overlook subtle patterns, contextual effects or interactions between multiple aspects of performance (Anderson & Sally, 2013; Memmert et al., 2017).

A data-driven approach enables a more **objective, systematic and reproducible evaluation of player performance**. Football performance can be described across multiple dimensions, including physical output (e.g. sprints, distance covered), technical actions (e.g. shots or passes), tactical context (e.g. formation and playing position), and behaviour under defensive pressure. Regardless of the analytical perspective adopted, football matches are ultimately decided by goals. Consequently, a fundamental problem in football analytics is estimating the probability that a given player will score a goal (Decroos et al., 2019).

You are therefore asked to build a **predictive model** to support coaches in better assessing a **player's likelihood of scoring a goal over the remaining duration of the game**. To facilitate this task, a base dataset has been prepared containing performance data for all players who participated in the **UEFA Under-21 European Championship held in Slovakia in 2025**. The dataset is structured to reflect the dynamic nature of a football match, with player performance observed in consecutive **15-minute intervals**, referred to as checkpoints.

The **base table** (`players_quarters_final`) is **ready-for-use** in your modelling, created based on selected sources mentioned below, and includes for **each player and each checkpoint** rolling performance aggregates describing activity in the most recent 15 minutes (variables prefixed with `last15_`) as well as cumulative aggregates summarizing activity from the start of the match up to the analysed moment (variables prefixed with `cumul_`). These aggregates capture, among other variables, information on shots taken and sprints performed. Importantly, the dataset also contains a **binary dependent**

1

 player_appearance_behaviour_under_pressure.csv 

 player_appearance_pass.csv 

 player_appearance_run.csv 

 player_appearance_shot_limited.csv 

 players_quarters_final.csv 



WARSAW econometric CHALLENGE

 ReSpo·Vision

WEC 2026 data description

Main dataset for the contest

players_quarters_final.csv
(3 486 rows) – player activity snapshots at match quarter-hour checkpoints

This is the main analytical dataset for the contest.

It was constructed by aggregating event-level data from selected source tables (see below) – `player_appearance_shot_limited.csv` and `player_appearance_run.csv` into a single player-level panel organised around fixed time checkpoints within each match.

The dataset covers 31 matches and 869 unique player appearances. Each row represents one player at one time checkpoint, capturing their activity both in the preceding 15-minute window and cumulatively from the start of the match up to that point. The binary target variable `scored_after` indicates whether the player scored a goal at any point after the checkpoint in the same match, excluding own goals.

Observation unit and time structure. Checkpoints are defined within match periods, with minutes counted locally from the start of each period. Up to seven checkpoints are possible per match: minute 15, 30, and 45 of the first half (`h1_15`, `h1_30`, `h1_45`), minute 15 and 30 of the second half (`h2_15`, `h2_30`), and – for matches that reached extra time – minute 45 of the second half (`h2_45`) and minute 15 of the first extra time period (`et1_15`). Only players who were actively on the pitch at the moment of the checkpoint are included, identified through the continuous `minute_in` and `minute_out` fields.

Feature groups. Variables with the prefix `last15_` describe the player's activity in the 15-minute window immediately preceding the checkpoint; variables with the prefix `cumul_` describe their accumulated activity from the start of the match. Both groups cover the same two domains: physical exertion (derived from `player_appearance_run.csv`) and shooting (derived from `player_appearance_shot_limited.csv`). Goals scored prior to the checkpoint were excluded from the shot features to prevent information leakage into the target variable.

Target variable. `scored_after` is a binary indicator equal to 1 if the player scored at least one non-own goal after the checkpoint in that match, and 0 otherwise. The target is heavily imbalanced: 203 positive cases out of 3,486 observations (approximately 5.8%), reflecting the natural rarity of goalscoring events in football.

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Detailed description of the challenge and the data



**All solutions have to be uploaded
using a form available on the website
<https://wec.wne.uw.edu.pl/>**

before 16:00 Sunday 26th of April 2026

GOOD LUCK!!!



LET THE COMPETITION BEGIN!